

From ‘also’ to ‘even’

Function alternations of the Mandarin particle *ye*

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Abstract This paper addresses a cross-linguistic puzzle concerning the frequent association between additive expressions and scalar interpretations. It focuses on the Mandarin additive particle *ye* and compares it with *dou*, which has a similar ‘even’-like use in *lian*-sentences despite differing substantially from *ye* in other constructions. I argue that the presuppositional properties of *ye*, including focus-sensitivity, additivity, and scalarity, derive from a single definedness condition, termed *Identity to Anti-Exclusion*. Parallel to the distributor-to-‘even’ alternation of *dou*, the ‘also’-to-‘even’ alternation of *ye* arises when exclusivity is defined in terms of likelihood rather than logical strength.

The paper further investigates the behavior of *ye* and *dou* across three complex scalar constructions. I argue that *ye* functions as a genuinely ‘even’-like scalar operator in minimizer-licensing constructions, but retains its identity as the non-scalar additive operator ‘also’ in evaluative statements and scalar concessives. By contrast, *dou*, which permits a non-additive scalar interpretation, patterns with *ye* in minimizer-licensing constructions but diverges from it in evaluative statements and scalar concessives.

Finally, the paper revisits the independence requirement on additive expressions, arguing that this requirement does not arise from lexical semantics, but rather from pragmatic constraints on the question-under-discussion.

Keywords: Multi-functional particles, additivity, scalarity, minimizers, evaluativity, concessives, Independence, Fluctuation, conditionals, exhaustification, Mandarin, cross-linguistic semantics, alternative semantics, dynamic semantics

1. Introduction

The Mandarin adverb *ye* exhibits both a simple additive use, as in (1), and a scalar additive use, as in (2), hereafter referred to as ‘also’ *ye* and ‘even’ *ye*, respectively. In both cases, *ye* associates with a focus and triggers an additive presupposition. The ‘also’ *ye* use has the same distribution as English *also*, while the ‘even’ *ye* use occurs primarily in *lian*-sentences, where the focus associate follows *lian* and precedes *ye*.

(1) ‘Also’ *ye*

- a. MALI *ye* lai-le.
Mary YE come-PERF
‘MARY_F also came.’
- b. Su *ye* qing-le MALI.
Sue YE invite-PERF Mary
‘Sue also invited MARY_F.’

(2) ‘Even’ *ye*

- a. Lian MALI *ye* lai-le.
LIAN Mary YE come-PERF
‘Even MARY_F came.’
- b. Su *lian* MALI *ye* qing-le.
Sue LIAN Mary YE invite-PERF
‘Sue even invited MARY_F.’

The alternation between the ‘also’ and ‘even’ uses is unlikely to be coincidental, as it is attested cross-linguistically across diverse language families (König, 1991; Forker, 2016). Parallel cases include Vietnamese *cũng*, Japanese *-mo*, Hindi *bhī*, Hmong *twb*, Romanian *și*, Greek *καί*, among many others. In a typological study on additive expressions conducted by Forker (2016), 34 of the 42 languages examined employ expressions that exhibit this ‘also’–‘even’ alternation.

Previous studies have identified several properties shared by ‘also’ and ‘even’, such as focus-sensitivity and additivity, and have recognized that their primary difference lies in the scalarity requirement (König, 1991, among others). However, it remains puzzling why additivity is so frequently accompanied by scalarity, rather than by some other property.

Mandarin is particularly well suited to investigating this puzzle because the ‘even’ use of *ye* overlaps that of another multi-functional Mandarin particle, *dou*. In contrast to *ye*, *dou* primarily functions as a universal distributor, left-associating with a non-atomic or divisible element and typically translated as ‘all’, as illustrated in (3). However, in *lian*-sentences, *dou* does not express distributivity; instead, it yields an ‘even’-like interpretation, as shown in (4).¹ These *lian ... dou* sentences closely parallel the *lian ... ye* sentences in (2), except that they do not trigger an additive presupposition: for instance, (4a) does not imply that someone other than Mary also came, in contrast to (2a).

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| <p>(3) Distributor <i>dou</i></p> <p>a. Tamen DOU lai-le.
they DOU come-PERF
‘They all came.’</p> <p>b. Sue (*<i>dou</i>) lai-le.
Sue DOU come-PERF
‘Sue *(all) came.’</p> | <p>(4) ‘Even’ <i>dou</i></p> <p>a. (Lian) MALI dou lai-le.
LIAN Mary DOU come-PERF
‘Even MARY_F came.’</p> <p>b. Su lian MALI dou qing-le.
Sue LIAN Mary DOU invite-PERF
‘Sue even invited MARY_F.’</p> |
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I argue that Mandarin *ye*, in both its ‘also’ and ‘even’ uses, is semantically vacuous except for triggering a definedness condition. I term this condition ‘*Identity to Anti-Exclusion*’: the asserted component of a *ye*-sentence must be contextually identical to the anti-exclusion of the additive component. Roughly, a sentence of the form ‘ ψ , *ye* ϕ ’ is felicitous only in contexts where ϕ is contextually identical to ‘not only ψ ’. The diagnostically presuppositional properties of *ye*, including focus-sensitivity, additivity, and scalarity, are all derived from this definedness condition.

I further argue that the ‘also’–‘even’ alternation of *ye* parallels the distributor–‘even’ alternation of *dou*, with both arising from variation in the definition of exclusivity. In particular, whether *ye* is interpreted as ‘also’ or ‘even’ depends solely on whether exclusivity is defined in terms of logical strength or likelihood: in the former case, exclusion targets non-entailed alternatives, whereas in the latter it targets less likely alternatives. This explanation echoes Xiang’s (2016, 2020) anti-exhaustifier analysis of *dou*, which likewise attributes the distributor–‘even’ alternation of *dou* to variation in the definition of exclusivity.

Beyond these core alternations, this paper further examines the behavior of *ye* and *dou*

¹The distributor–‘even’ alternation is also seen with Vietnamese *cũng* and Hmong *puav* (Beryl Bui p.c.).

across three related scalar constructions: minimizer-licensing sentences, evaluative statements, and scalar concessives (see (5)–(7)). I argue that, despite all three constructions involving a scalar component, only in the construction in (5), where *ye* associates with a minimizer, *ye* functions as ‘even’. I further argue that the similarities between *ye* and *dou* in minimizer-licensing sentences stem from their shared scalar inference, whereas their differences in evaluative statements and scalar concessives arise from a contrast in additivity.

- (5) Mali (lian) YI-ge-ren dou/ye mei qing.
 Mary LIAN one-CL-person DOU/YE NEG invite
 ‘Mary didn’t even invite a SINGLE person.’ (Minimizer-association)
- (6) Mali dou/ye dadao ZHONGJI le.
 Mary DOU/YE achieve mid.level LE
 ‘Mary has already achieved the MID-level.’
 $\rightsquigarrow$ *Mary is competitive/ somewhat competitive.* (Evaluative statements)
- (7) Jiusuan zhe-ben-shu youdian-gui, wo ye/?dou yao mai.
 JIUSUAN this-CL-book a.little-expensive, I YE/DOU will buy
 ‘Even if this book is a little expensive, I would buy it.’ (Scalar concessives)

It should be noted that this paper does not address cases where *ye* associates with a free-choice item (FCI) or a negative polarity item (NPI), since a full account of such constructions would require a discussion in parallel to the free-choice (FC-)trigger use of *dou*, rather than its distributor or ‘even’ uses, and would involve a different set of linguistic constraints.

The rest of the paper is organized as follows. Sections 2 and 3 present an analysis that uniformly explains the basic ‘also’ and ‘even’ interpretations of *ye*, and predicts the parallel function alternations of *dou* and *ye*. Sections 4 to 6 compare the behavior of *ye* and *dou* across three complex scalar constructions, including minimizer-licensing sentences, evaluative statements, and scalar concessives. Section 7 addresses a remaining issue concerning independence. Section 8 concludes.

2. Explain ‘also’ *ye*

2.1. In a nutshell

‘Also’ *ye* behaves like the English preverbal additive particle *also*. It freely associates with focus in the preadjacent and triggers an additive presupposition.

- (8) a. MALI ye qing-le Su.
 Mary YE invite-PERF Sue
 ‘MARY_F also invited Sue.’ $\rightsquigarrow$ *Someone other than Mary invited Sue.*
- b. Mali ye qing-le SU.
 Mary YE invite-PERF Sue
 ‘Mary also invited SUE_F.’ $\rightsquigarrow$ *Mary invited someone other than Sue.*

In a nutshell, I argue that an ‘also’ *ye*-sentence is subject to a definedness condition that I call *Identity to Anti-exclusion (IAE)*. Under this condition, the asserted component must be contextually identical to the anti-exhaustification of the additive component. For a *ye*-sentence of the form $ye_{\psi}(\phi)$, where ϕ is the asserted component and ψ the additive component, IAE requires that ϕ is contextually equivalent to ‘(relative to ϕ) not only ψ ’.² The ‘not only’ component, in turn, introduces focus-sensitivity and presupposes the truth of the additive component. For example, suppose there are only two relevant individuals, John and Mary, and it is part of the common ground that Mary came. In this context, the sentence *John also came* is felicitous because asserting *John came* has the same discourse effect as asserting *Not only Mary came*.

The IAE requirement can be formally captured within Heim’s (1982, 1983) model of dynamic semantics. I first introduce the basics of this framework and then present a formal account of the ‘also’ use of *ye*.

2.2. Basics of Heim’s model of dynamic semantics

Heim (1982, 1983) defines sentence meaning dynamically, in terms of its contribution to the discourse, called *context-change (cc-)potential*. The cc-potential of a sentence is a (possibly partial) function of type $\langle st, st \rangle$; it applies to the context in which the sentence is uttered, namely, the set of worlds compatible with the mutual beliefs of the interlocutors, and returns an updated context.

As schematized in (9), the cc-potential of an atomic sentence ϕ is defined only if the input context entails the presuppositions of ϕ . When defined, updating the context with ϕ amounts to restricting the context to those worlds in which ϕ is true.³

(9) For any closed atomic sentence ϕ :

$$c + \phi = \begin{cases} c \cap \llbracket \phi \rrbracket & \text{if } c \subseteq \text{PRES}(\phi) \\ \text{undefined} & \text{otherwise} \end{cases}$$

(10) For any ϕ , p is a presupposition of ϕ iff $c + \phi$ is defined only if $c \subseteq p$.

For example, $c + \text{Mary’s daughter came}$ is defined only if *Mary has a daughter* is true in every world in c . When defined, $c + \text{Mary’s daughter came}$ returns the subset of worlds in c in which

²The idea of deriving additivity from anti-exhaustification has also been proposed by Nicolae (2020) and Fălăuş & Nicolae (2022). In their account of the Romanian additive/scalar particle *şi*, they argue that $\llbracket \text{şi}(p) \rrbracket$ implicates the exclusion of the pre-exhaustified alternative $\text{EXH}(p)$, as illustrated in (i). However, because this approach treats the additive inference as a quantity-based implicature, it does not straightforwardly capture the presuppositional status of additivity.

(i) $\llbracket \text{şi} \rrbracket(p) = \text{EXH}_{\text{ADD}}(\text{EXH}(p)) = p \wedge \neg \text{EXH}(p) = p \wedge \neg(p \wedge \neg q) = p \wedge q$

³Notations:

- (i) a. c : a context (set), the set of worlds compatible with mutual belief
- b. ϕ : the logical form of a sentence
- c. $+$: the context-incrementation operation, which combines a context with a sentence
- d. $\llbracket \phi \rrbracket$: the set of worlds where ϕ is true
- e. $\text{PRES}(\phi)$: the set of worlds where the presuppositions of ϕ are true

Mary's daughter came.

This framework captures the distinct discourse statuses of assertions and presuppositions: assertions contribute new information to the ongoing discourse, whereas presuppositions are already assumed by the context.

Heim's model is particularly well-known for its ability to account for the projection behavior of presuppositions under boolean connectives. For present purposes, it suffices to consider negation. As schematized in (11), updating a context with a negated sentence *not-φ* proceeds in two steps: first compute $\phi + c$, and then subtract the resulting set from c . This definition captures the projection of presuppositions under negation, since $c + \text{not-}\phi$ is defined only if $c + \phi$ is defined.

$$(11) \quad \text{Negation rule} \\ c + \text{not-}\phi = c - (c + \phi)$$

Compared to alternative approaches, a key advantage of Heim's model for the present analysis is that the context-incrementation operation '+' applies to the full structural representation of an expression at Logical Form, not just a "black box" of a string. As a result, it has access to the internal structure of a complex expression. For example, when applied to a negative sentence *not-φ*, the system recognizes both the negation and its embedded clause. This property makes it possible to compare the cc-potentials of expressions with shared subparts, which will be crucial for formalizing the IAE constraint.

2.3. Formal analysis

2.3.1. Proposal

Assuming Heim's (1982, 1983) model, I formalize the IAE requirement as an identity in context incrementation, as stated in (12).

$$(12) \quad \text{The Identity to Anti-exclusion (IAE) Constraint} \\ \text{For any context } c \text{ and sentences } \phi \text{ and } \psi, c + ye_{\psi}(\phi) \text{ is defined only if:} \\ c + \phi = c + \text{not-ONLY}_{\phi}(\psi)$$

The cc-potential definition in (13) has two assumptions. First, 'also' *ye* does not contribute new at-issue content; for instance, *John also came* simply asserts that John came. Second, *ye* introduces the IAE requirement, which, as argued below, derives both focus-sensitivity and additivity.

$$(13) \quad \text{Semantics of 'also' } ye \\ \text{For any context } c \text{ and sentences } \phi \text{ and } \psi, \text{ we have:} \\ c + ye_{\psi}^{\text{ALSO}}(\phi) = \begin{cases} c + \phi & \text{if } c + \phi = c + \text{not-ONLY}_{\phi}(\psi) \\ \text{undefined} & \text{otherwise} \end{cases}$$

Within the IAE condition, the operator *ONLY* has roughly the same semantics as the exclusive particle *only*: it presupposes the truth of its prejacent and excludes non-entailed alternative(s) (cf. Horn, 1969). However, a key difference is that the exclusion operation

in $ONLY_\phi$ targets not a full set of focus-alternatives, but rather a single focus-containing sentence, namely, the asserted component of ye . A cc-potential definition and the corresponding truth-conditional definition are given in (14).

(14) **Logical-strength-based $ONLY$**

- a. For any context c and sentences ϕ and ψ :

$$c + ONLY_\phi(\psi) = \begin{cases} c + not-\phi & \text{if } \psi \in F\text{-ALT}(\phi), c \subseteq \llbracket \psi \rrbracket, \text{ and } \psi \not\Rightarrow \phi \\ c & \text{if } \psi \in F\text{-ALT}(\phi), c \subseteq \llbracket \psi \rrbracket, \text{ and } \psi \Rightarrow \phi \\ \text{undefined} & \text{otherwise} \end{cases}$$

($c + ONLY_\phi(\psi)$ is defined only if ψ is a focus-alternative of ϕ and c entails ψ ; when defined, updating c with $ONLY_\phi(\psi)$ amounts to excluding ϕ unless it is already entailed by ψ .)

- b. $\llbracket ONLY_\phi(\psi) \rrbracket = \lambda w : \psi \in F\text{-ALT}(\phi) \wedge \llbracket \psi \rrbracket^w = 1. (\psi \Rightarrow \phi) \vee (\llbracket \phi \rrbracket^w = 0)$

($ONLY_\phi(\psi)$ is defined in w only if ψ is a focus-alternative of ϕ and ψ is true in w ; when defined, $ONLY_\phi(\psi)$ is true in w iff either ψ entails ϕ or ϕ is false in w .)

The definition in (13) covers only cases with one additive component. In cases with multiple additive components, the IAE condition applies point-wise, as defined in (15).

- (15) For any context c , sentence ϕ , and sequence of sentences $\vec{\psi}$:

$$c + ye_{\vec{\psi}}^{\text{ALSO}}(\phi) = \begin{cases} c + \phi & \text{if } \forall \psi \in \vec{\psi} : c + \phi = c + not-ONLY_\phi(\psi) \\ \text{undefined} & \text{otherwise} \end{cases}$$

2.3.2. Predictions

The IAE equation—namely, $c + \phi = c + not-ONLY_\phi(\psi)$ —holds for c if and only if, whenever the left-hand side ($c + \phi$) is defined, the right-hand side ($c + not-ONLY_\phi(\psi)$) is also defined and yields the same context set.

The computation in (16) shows that $c + not-ONLY_\phi(\psi)$ is defined if and only if c satisfies the presuppositions triggered by $ONLY$, namely, focus-sensitivity and the truth of the additive component ψ .⁴

- (16) $c + not-ONLY_\phi(\psi)$ is defined,

⁴This is where Heim's (1982, 1983) context-change model proves particularly useful: the two presuppositional properties of 'also' emerge as definedness conditions of an intermediate step that involve the presupposition trigger $ONLY$. By contrast, if we adopt the standard textbook approach of encoding presuppositions as definedness conditions of a partial function, using the form ' $\lambda v : \beta.\alpha$ ' (Heim & Kratzer, 1998), the proposed semantics of ye could at best be represented as in (i).

$$(i) \quad \llbracket ye_{\vec{\psi}}^{\text{ALSO}}(\phi) \rrbracket = \lambda w. [\llbracket \phi \rrbracket^w = 1 \leftrightarrow \llbracket ONLY_\phi(\psi) \rrbracket^w = 0]. \llbracket \phi \rrbracket^w = 1$$

However, this representation is technically inadequate, as the lambda notation does not permit the definedness condition itself to be trivalent or otherwise undefined. As a result, it cannot properly capture the intermediate presuppositional step required by the IAE condition.

An alternative framework to handling such 'nested presuppositions' is multi-dimensional semantics (Karttunen & Peters, 1977, 1979). However, this framework offers no clear advantage over the present analysis and does not straightforwardly capture the identity condition central to IAE.

- a. iff $c + \text{ONLY}_\phi(\psi)$ is defined; by negation rule (11)
- b. iff $c \subseteq \text{PRES}(\text{ONLY}_\phi(\psi))$;
- c. iff $\psi \in \text{F-ALT}(\phi)$ and $c \subseteq \llbracket \psi \rrbracket$. by def of *ONLY* (14a)

Next, when both sides of the IAE equation are defined, this equation holds non-trivially if and only if $\psi \not\Rightarrow \phi$. A formal proof is given in (17): when $\psi \not\Rightarrow \phi$, the IAE equation holds; when $\psi \Rightarrow \phi$, the equation fails unless both ϕ and ψ are contradictions.

(17) Suppose $c + \phi$ and $c + \text{not-ONLY}_\phi(\psi)$ are both defined. Then:

- a. i. If $\psi \not\Rightarrow \phi$:
 - ii. $c + \text{not-ONLY}_\phi(\psi)$

$$= c - (c + \text{ONLY}_\phi(\psi))$$
 by negation rule (11)

$$= c - (c + \text{not-}\phi)$$
 by (17a-i) and def of *ONLY* (14a)

$$= c - (c - (c + \phi))$$
 by negation rule (11)

$$= c + \phi$$
- b. i. If $\psi \Rightarrow \phi$:
 - ii. $c + \text{not-ONLY}_\phi(\psi)$

$$= c - (c + \text{ONLY}_\phi(\psi))$$
 by negation rule (11)

$$= c - c$$
 by (17b-i) and def of *ONLY* (14a)

$$= \emptyset$$

The same result follows from the truth-conditional definition of *ONLY* in (14b). When defined, '*not-ONLY* _{ϕ} ψ ' is true iff $\psi \not\Rightarrow \phi$ and ϕ is true. Thus, assuming ϕ is true, a sentence of the form '*not ONLY* _{ϕ} ψ ', once defined, is true if and only if ψ does not entail ϕ .

In sum, the cc-potential definition in (13) is effectively equivalent to (18). The IAE requirement is satisfied if and only if the additive component is a true focus-alternative that does not entail the asserted component, thereby deriving both focus-sensitivity and additivity.

(18) $c + ye_\psi^{\text{ALSO}}(\phi)$ is defined only if

- a. $\psi \in \text{F-ALT}(\phi)$ focus-sensitivity
- b. $c \subseteq \llbracket \psi \rrbracket$ and $\psi \not\Rightarrow \phi$ additivity

when (a) and (b) are satisfied, $c + ye_\psi^{\text{ALSO}}(\phi) = c + \phi$.

At this point, the analysis yields results equivalent to standard accounts of additive particles (e.g., König 1991): *ye* (like *also*) asserts the truth of the prejacent and presupposes the truth of a non-stronger alternative. Given this equivalence, what is gained by deriving these presuppositional properties via an intermediate comparison with a 'not only' clause?

The answer lies in the central puzzle of this paper: the 'also'–'even' alternation. Cross-linguistically, additivity is frequently accompanied with scalarity. As I argue in the following section, this connection is not accidental, but follows from variation in the definition of 'only'.

3. Explain 'even' *ye*

Despite clear distinctions in their other uses, *ye* and *dou* are strikingly similar in their 'even' use. In what follows, I argue that this similarity is not accidental: the 'also'-to-'even' shift of

ye closely parallels the distributor-to-‘even’ shift of *dou*.

This section begins by introducing Xiang’s (2016, 2020) anti-exhaustifier analysis of *dou*. Under this analysis, *dou* primarily functions as a distributor when exclusivity is defined in terms of logical strength, which is taken to be the default. However, it acquires an ‘even’ interpretation when exclusivity is instead defined in terms of likelihood.

I show that incorporating this idea into the proposed context-change analysis of *ye* provides a straightforward account of its ‘even’ use: when exclusivity likelihood-based, the IAE requirement gives rise not only to focus-sensitivity and additivity, but also to scalarity.

3.1. From distributor *dou* to ‘even’ *dou*

3.1.1. Observations

The Mandarin particle *dou* is highly multi-functional. Most notably, it can function as a universal distributor (‘both/all’), a scalar operator (‘even’), or an associate of FCIs or NPIs. In addition, *dou* may give rise to semantic effects such as evaluativity, maximality, and the removal of homogeneity. This section focuses on the contrast between its distributor and the ‘even’ uses.

In its distributor use, *dou* distributes over the subparts of a plural or divisible element appearing to its left. In sentence (19a), although the bare noun *fangzi* ‘house’ is number-unmarked, the sentence rejects a collective reading in which the individuals jointly bought one house. By contrast, in its ‘even’ use, *dou* is not subject to a plurality or distributivity requirement. As seen in (19b), ‘even’ *dou* is compatible with collective contexts and may associate with atomic elements. Instead, it introduces a scalar inference that the asserted proposition is relatively unlikely. Notably, the scalar marker *lian* in (19b) is optional, and the two readings can be disambiguated prosodically: the associate of distributor *dou* is usually unstressed, whereas the associate of ‘even’ *dou* must bear stress.

- (19) a. Tamen/*Mali dou/DOU mai-le fangzi.
They/Mary DOU/DOU buy-PERF house
‘They all bought houses.’/*‘Mary all bought houses.’ (Distributor)
- b. (Lian) Tamen/MALI dou mai-le fangzi.
LIAN they/Mary DOU buy-PERF house
‘Even THEY/Mary_F bought houses/ a house.’ (‘Even’)

A further difference is that ‘even’ *dou* cannot associate with quantificational determiners expressing universal force or large quantity, as shown in (20a,b). When *lian* precedes an ‘all/every/many’-NP, *dou* must instead associate into a smaller, non-quantificational element, as in (20c). This pattern parallels the behavior of English *even*, as reflected in the translations. This restriction is related to the independence requirement of ‘even’, which will be addressed in section 7.

- (20) a. Suoyou/henduo-ren dou tongguo-le kaoshi.
all/many-person DOU pass-PERF exam
‘All/Many people passed the exam.’ (Distributor)

- b. * Lian SUOYOU/HENDUO-ren dou tongguo-le kaoshi.
 LIAN all/many-person DOU pass-PERF exam
 ‘*Even ALL/MANY_F people passed the exam.’ (‘Even’)
- c. Lian henduo CHUXUEzhe dou tongguo-le kaoshi.
 LIAN many beginner DOU passed-PERF exam
 ‘Even many beGINners_F passed the exam.’ (‘Even’)

3.1.2. *Dou* as an anti-exhaustifier (Xiang 2016, 2020)

Xiang (2016, 2020) accounts for the function alternations of *dou* under a uniform analysis by treating *dou* as an *anti-exhaustifier*. Under this view, *dou* asserts not only the prejacent but also the anti-exhaustification of certain alternatives.⁵ Briefly, a sentence of the form ‘*dou* ϕ ’ roughly conveys ‘ ϕ and not only ψ ’, where ψ is what Xiang terms as a *sub-alternative* of ϕ . The exact definition of sub-alternatives varies by the definition of exclusivity, and this variation gives rise to different uses of *dou*.

A formal definition is given in (21): *dou* affirms the prejacent and negates the exhaustification of each sub-alternative, and is defined only if at least one such sub-alternative exists.

- (21) **Semantic of *dou*** (Xiang, 2016, 2020)
- $$\llbracket dou_C \rrbracket = \lambda p \lambda w : \underbrace{SUB(p, C) \neq \emptyset}_{\text{non-vacuity}} \cdot \underbrace{p(w) = 1}_{\text{prejacent}} \wedge \underbrace{\forall q \in SUB(p, C)[O_C(q)(w) = 0]}_{\text{anti-exclusion}}$$
- a. *Non-vacuity presupposition*: The prejacent has at least one sub-alternative.
 b. *Prejacent inference*: The prejacent is true.
 c. *Anti-exclusion inference*: The exhaustification of each sub-alternative is false.

Within the asserted anti-exclusion inference, the *O*-operator functions similarly to the exclusive particle *only*, except that it asserts, rather than presupposes, the truth of its prejacent (Chierchia et al., 2012).

- (22) $O_C := \lambda q \lambda w : q(w) = 1 \wedge \forall r \in EXCL(p, C)[r(w) = 0]$
 (For any q , $O_C(q)$ is true in w iff q is true in w and all excludable alternatives of q in C are false in w .)

The notion of *sub-alternative* is defined in contrast to *excludable alternative*: it refers to any alternative that cannot be excluded under the given definition of exclusivity, excluding the prejacent itself as well as any alternatives that are equally ranked as the prejacent.

Next, Xiang argues that which interpretation *dou* obtains is entirely determined by how exclusivity is defined. As summarized in Table 1, different definitions of exclusion yield different sets of sub-alternatives—such as logically weaker alternatives, more likely alternatives,

⁵Xiang (2016, 2020) calls *dou* a ‘pre-exhaustification exhaustifier’, due to influence from Chierchia’s (2006) analysis of FC. However, Xiang (2022) modifies the name to ‘anti-exhaustifier’: a pre-exhaustification exhaustifier is an exhaustifier targets pre-exhaustified alternatives, while *dou*, as an anti-exhaustification operator, targets sub-alternatives that are not pre-exhaustified.

or non-innocently-excludable alternatives—and thereby give rise to distinct functions of *dou*. In what follows, I focus on two such uses: the distributive use and the ‘even’ use.

Base of exclusivity	logical strength	likelihood	innocent excludability	...
Sub-alternative	weaker	more likely	non-innocently-excludable	...
Use of <i>dou</i>	distributor	‘even’	FC-trigger	...

Table 1: Variation in the definition of exclusion determines the function of *dou*

Distributor *dou* The distributor use arises when exclusivity is defined, by default, in terms of logical strength, and thus constitutes the primary use of *dou*. Under this definition, excludable alternatives are those not entailed by the prejacent, while sub-alternatives are those that are logically weaker than the prejacent.

- (23) a. $\text{EXCL}(p, C) = \{q \mid q \in C \wedge p \not\subseteq q\}$ (based on logical strength)
 (The set of alternatives that are not entailed by p)
 b. $\text{SUB}(p, C) = \{q \mid q \in C \wedge p \subset q\}$
 (The set of alternatives that are weaker than p)
 c. $O_C = \lambda q \lambda w : q(w) = 1 \wedge \forall r \in C [r(w) = 1 \rightarrow q \subseteq r]$
 (q is true, and q entails all true alternatives in C .)

Under this configuration, the anti-exclusion inference is entailed by the prejacent inference and is therefore semantically vacuous. What remains is the non-vacuity presupposition, which yields the distributivity/plurality requirement. In (24), the prejacent must receive a distributive interpretation, since otherwise no sub-alternative would be available

- (24) Andy and Billy *dou* bought *fangzi*.
 a. $[dou_C [\phi \text{ [Andy and Billy]}_F \text{ bought a house }]]$ (Distributive)
 b. $\llbracket \phi \rrbracket = f(a) \wedge f(b)$
 c. $\text{SUB}(\llbracket \phi \rrbracket, C) = \{f(a), f(b)\}$
 d. $\llbracket dou_C(\phi) \rrbracket = [f(a) \wedge f(b)] \wedge \neg O_C(f(a)) \wedge \neg O_C(f(b)) = f(a \oplus b)$

‘Even’ *dou* The ‘even’-like interpretation likewise follows from the anti-exhaustifier analysis. When the ordering of alternatives shifts from logical strength to likelihood, sub-alternatives are defined as those alternatives that are more likely than the prejacent.⁶ Consequently, the

⁶Xiang (2020) does not explicitly specify the sense in which one proposition is said to be less likely than another. For a sentence such as *Even JOHN passed the test*, in what sense is the proposition ‘John passed the test’ considered relatively unlikely, given that it is known to be true? This note clarifies that likelihood should be understood in terms of what Kratzer (1981) calls *Human Possibility*. On this view, likelihood is not a measure of actual truth, but is defined relative to a stereotypical ordering source representing an ideal or normal state of affairs. It tracks normality rather than actuality: a proposition is considered unlikely if it is false in the best-ranked worlds, even if it happens to be true in the actual world. Some core definitions are given below.

- (i) Given an anchoring world w and an ordering source o , we have:
 a. $o(w)$: the set of propositions describing the expected course of events in w .

exhaustifier involved in the anti-exclusion inference shifts from the standard exhaustifier O to a likelihood-based scalar exhaustifier, denoted as O^L .

- (25) a. $\text{EXCL}(p, C) = \{q \mid q \in C \wedge p >_{\text{likely}} q\}$ (based on likelihood)
 (The set of alternatives that less likely than p)
 b. $\text{SUB}(p, C) = \{q \mid q \in C \wedge p <_{\text{likely}} q\}$
 (The set of alternatives that are more likely than p)
 c. $O_C^L = \lambda q \lambda w : q(w) = 1 \wedge \forall r \in C [r(w) = 1 \rightarrow q \leq_{\text{likely}} r]$
 (q is true, and q is the least likely among the true alternatives in C .)

Under this shift, *dou* becomes semantically equivalent to (non-additive) *even*: its non-vacuity presupposition corresponds to the scalar presupposition of *even*, while the asserted anti-exclusion component is again vacuous.⁷

- (26) $\llbracket \text{dou}_C \rrbracket$
 $= \lambda p \lambda w : \exists q \in \text{SUB}(p, C). p(w) = 1 \wedge \forall q \in \text{SUB}(p, C) [O_C^L(q)(w) = 0]$
 $= \lambda p \lambda w : \exists q \in \text{SUB}(p, C).$
 $\quad [p(w) = 1 \wedge \forall q \in \text{SUB}(p, C) \exists r \in C [r(w) = 1 \wedge q >_{\text{likely}} r]]$
 $= \lambda p \lambda w : \exists q \in C [q >_{\text{likely}} p].$
 $\quad [p(w) = 1 \wedge \forall q \in C [q >_{\text{likely}} p \rightarrow \exists r \in C [r(w) = 1 \wedge q >_{\text{likely}} r]]]$
 $= \lambda p \lambda w : \exists q \in C [q >_{\text{likely}} p]. p(w) = 1$

In sum, by analyzing *dou* as an anti-exhaustifier, Xiang predicts that its distinct lexical meanings arise from minimal variation in the definition of exclusivity. When exclusivity is defined in terms of logical strength, *dou* is truth-conditionally vacuous but presupposes the existence of a logically weaker alternative, thereby functioning as a distributor. When exclusivity is defined in terms of likelihood, *dou* remains truth-conditionally vacuous but presupposes the existence of a more likely alternative, yielding an ‘even’-like interpretation.⁸

In contrast to competing approaches that also aim to derive the various functions of *dou* uniformly, Xiang’s anti-exhaustifier analysis does not attribute the apparent multi-functionality

-
- b. For two worlds u and v accessible from w , u is ranked higher than v relative to $o(w)$ iff u satisfies more of the propositions in $o(w)$ than v does. Formally:
 $u <_{o(w)} v$ iff $\{p \in o(w) : v \in p\} \subset \{p \in o(w) : u \in p\}$
 c. For two propositions ψ_p and q , q is strictly more likely than ψ_p iff (i) for every p -world u , there is a q -world v s.t. v is ranked at least as good as u ; and (ii) there is a q -world v' that is ranked strictly better than all p -worlds. Formally:
 $q >_{\text{likely}} p$ iff (i) $\forall u \in p \exists v \in q [v \leq_{o(w)} u]$ and (ii) $\exists v' \in q \forall u \in p [v' <_{o(w)} u]$

Under this definition, the distributor-to-‘even’ alternation of *dou* can be understood as a semantic weakening. If p entails q , then q is guaranteed to be at least as likely as p : for every p -world u , there is q -world (namely, u itself) that is ranked equally high.

⁷On why the asserted anti-exclusion inference is vacuous: when p is true, any alternative of p that is more likely than p is less likely than some true alternative r , where $r = p$.

⁸To be clear, *dou* is not always truth-conditionally vacuous. In the FC-trigger use, obtained when exclusion is defined in terms of innocent excludability, the anti-exclusion inference asserted by *dou* is not entailed by the preadjacent and is not semantically vacuous.

to a weak or under-specified lexical meaning. Instead, Xiang treats the stronger meaning, namely, the one based on logical strength and yielding the distributive use, as the primary meaning of *dou*. The underlying motivation is cognitive: logical strength is more readily accessible than alternative concepts such as likelihood. This view is further supported by diachronic evidence showing that the distributor use emerges earlier historically than other uses.

3.2. Observations: ‘even’ *ye* versus ‘even’ *dou*

As noted, *ye* and *dou* are largely alike in their ‘even’ uses. For both particles, the ‘even’ interpretation arises in *lian*-sentences, where *dou* or *ye* associates with (or into) the argument of *lian*, as illustrated in (27). These sentences carry a scalar presupposition that the asserted prejacent is less expected or less likely than the relevant alternatives.

- (27) a. Lian MALI *dou/ye lai-le*.
 LIAN Mary DOU/YE come-PERF
 ‘Even MARY_F came.’
 ~> *Compared to the others, it is unlikely that Mary would come.*
- b. Lian Mali-de NAI_{nai} *dou/ye lai-le*.
 LIAN Mary-POSS grandma DOU/YE come-PERF
 ‘Even Mary’s GRANDma_F came.’
 ~> *Compared to the others, it is unlikely that anyone’s GRANDma would come.*

The parallel between ‘even’ *ye* and ‘even’ *dou* is further reflected in their largely identical behavior in association with minimizers, as shown in (28). In such cases, *lian* may be omitted.

- (28) a. Mali (lian) YI-ge-ren *dou/ye mei qing*.
 Mary LIAN one-CL-person DOU/YE NEG invite
 ‘Mary didn’t even invite a SINGLE person.’
- b. Jintian (lian) YI-ge-ren *dou/ye mei lai*.
 today LIAN one-CL-person DOU/YE NEG come
 ‘Today not even a SINGLE person came.’

Despite these similarities, there are two subtle distinctions between *ye* and *dou* in their ‘even’ uses. First, unlike ‘even’ *dou*, ‘even’ *ye* triggers an additive presupposition (Liu, 2023). In example (29), *ye* is infelicitous because it presupposes not only the existence of a more likely alternative of the form ‘*x* came,’ but also the truth of such an alternative. While it is difficult to determine whether English *even* is inherently additive, Mandarin morphologically distinguishes between additive and non-additive ‘even’. These data further show that additivity is not a prerequisite for scalarity.

- (29) Lian MALI *dou/#ye lai-le*, zenme qita-ren *dou mei lai*?
 LIAN Mary DOU/YE come-PERF, how other-person DOU_{ALL} NEG come
 ‘Even MARY_F came, how come that all the others didn’t come?’

Second, ‘even’ *ye* is more dependent on the presence of *lian*. All else being equal, omitting *lian* does not affect the ‘even’ use of *dou*, but it may make it make an ‘even’ interpretation for *ye* less accessible, as illustrated in (30).

- (30) a. MALI *dou/ye lai-le*.
 Mary DOU/YE come-PERF
 Preferred with *dou*: ‘Even MARY_F came.’
 Preferred with *ye*: ‘MARY_F also came.’
 b. Su MALI *dou/ye qing-le*.
 Sue Mary DOU/YE invite-PERF
 Preferred with *dou*: ‘Sue even invited MARY_F.’
 Preferred with *ye*: ‘As for MARY, Sue as well invited her.’

It is straightforward why *dou* in (30) is unambiguously interpreted as ‘even’: distributor *dou* requires its associate to be non-atomic and de-accented (see (19a)), whereas in (30) the associate is atomic and bears stress. Why, then, does *ye* in the same sentences favor the ‘also’ interpretation? In principle, the ‘even’ use of *ye* should also be available without overt *lian*: as shown in (28), omitting *lian* does not affect the minimizer-association use of either *dou* or *ye*.

In what follows, I present a formal account of ‘even’ *ye*, and then explain its subtle differences from ‘even’ *dou*.

3.3. From ‘also’ *ye* to ‘even’ *ye*

It may be tempting to analyze *ye* as a simple additive operator and derive scalarity independently from *lian*. This approach is common in the literature (with the exception of Liu 2023). However, such analyses face two problems. First, they fail to predict the distributor–‘even’ alternation of *dou*: if scalarity were contributed solely by *lian*, we would expect *lian ... dou* to exhibit both distributive and scalar effects, contrary to the observations in (19). Second, they do not capture the cross-linguistic ‘also’–‘even’ alternation, since they attribute scalarity to a separate particle rather than to the additive expression itself. Thus, regardless of the role of *lian*, scalarity must be derivable from the lexical semantics of *dou* and *ye*.

As argued by Xiang (2016, 2020), when exclusivity is defined in terms of likelihood, *dou* loses its distributive requirement and obtains a scalar presupposition, yielding an ‘even’ interpretation. I argue that *ye* undergoes a parallel shift: when exclusivity is likelihood-based, *ye* likewise gives rise to scalarity, while retaining its additive presupposition.

The cc-potential definition of ‘even’ *ye* is given in (31). It differs from that of ‘also’ *ye* in (13) only in that *ONLY* is replaced by its likelihood-based counterpart *ONLY*^L.

- (31) **Semantics of ‘even’ *ye***
 For any context *c* and sentences ϕ and ψ :

$$c + ye_{\psi}^{\text{EVEN}}(\phi) = \begin{cases} c + \phi & \text{if } c + \phi = c + \text{not-ONLY}_{\phi}^L(\psi) \\ \text{undefined} & \text{otherwise} \end{cases}$$

The *ONLY*^L-operator corresponds to the likelihood-based exhaustifier *O*^L (see (25)), except

that it presupposes rather than asserts the truth of its argument. Its cc-potential and truth-conditional definitions are given in (32), parallel to those of *ONLY* in (14).

(32) **Likelihood-based *ONLY*^L**

a. For any context c and two sentences ϕ and ψ , we have:

$$c + \text{ONLY}_{\phi}^L(\psi) = \begin{cases} c + \text{not-}\phi & \text{if } \psi \in \text{F-ALT}(\phi), c \subseteq \llbracket \psi \rrbracket, \text{ and } \psi >_{c\text{-likely}} \phi \\ c & \text{if } \psi \in \text{F-ALT}(\phi), c \subseteq \llbracket \psi \rrbracket, \text{ and } \psi \leq_{c\text{-likely}} \phi \\ \text{undefined} & \text{otherwise} \end{cases}$$

($c + \text{ONLY}_{\phi}^L(\psi)$ is defined only if ϕ is an alternative of ψ and c entails ψ ; where defined, updating c with $\text{ONLY}_{\phi}^L(\psi)$ is like denying ϕ in c if ϕ is contextually less likely than ψ and is vacuous otherwise.)

b. $\llbracket \text{ONLY}_{\phi}^L(\psi) \rrbracket = \lambda w : \psi \in \text{F-ALT}(\phi) \wedge \llbracket \psi \rrbracket^w = 1. (\psi \leq_{c\text{-likely}} \phi) \vee (\llbracket \phi \rrbracket^w = 0)$
($\text{ONLY}_{\phi}^L(\psi)$ is defined in w only if $\psi \in \text{F-ALT}(\phi)$ and ψ is true in w ; when defined, $\text{ONLY}_{\phi}^L(\psi)$ is true in w iff ψ is not contextually more likely than ϕ or ϕ is false in w .)

As with ‘also’ *ye*, the IAE requirement derives focus-sensitivity and the truth of the additive component: $c + \text{not-ONLY}_{\phi}^L(\psi)$ is defined only if c entails the presuppositions triggered by ONLY^L . Crucially, however, the identity condition now imposes a stronger requirement. Unlike the logical-strength case, the equivalence between $c + \phi$ and $c + \text{not-ONLY}_{\phi}^L(\psi)$ is not guaranteed merely by $\psi \not\Rightarrow \phi$. Specifically:

(33) When $\psi \in \text{F-ALT}(\phi)$ and $c \subseteq \psi$,

- a. $c + \phi = c + \text{not-ONLY}_{\phi}^L(\psi) + \phi$ iff $\psi \not\Rightarrow \phi$;
- b. $c + \phi = c + \text{not-ONLY}_{\phi}^L(\psi) + \phi$ iff $\psi >_{c\text{-likely}} \phi$.

Note that the likelihood-based requirement $\psi >_{c\text{-likely}} \phi$ is stronger than the logical strength-based requirement $\psi \not\Rightarrow \phi$; for example, two logically equivalent propositions may be contextually equally likely. The proof below for (33b) parallels the proof in (17) for (33a):

(34) a. i. If $\psi >_{c\text{-likely}} \phi$: ii. $c + \text{not-ONLY}_{\phi}^L(\psi)$ $= c - (c + \text{ONLY}_{\phi}^L(\psi))$ $= c - (c + \text{not-}\phi)$ $= c - (c - (c + \phi))$ $= c + \phi$	b. i. If $\psi \leq_{c\text{-likely}} \phi$: ii. $c + \text{not-ONLY}_{\phi}^L(\psi)$ $= c - (c + \text{ONLY}_{\phi}^L(\psi))$ $= c - c$ $= \emptyset$
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In short, the cc-potential definition of ‘even’ *ye* in (31) is effectively equivalent to (35). Compared to the logical-strength-based semantics in (18), this likelihood-based semantics presupposes not only focus-sensitivity and additivity, but also a scalar inference that the asserted component is contextually less likely than the additive component, thereby yielding an additive ‘even’ interpretation. Crucially, this scalarity condition is not introduced independently, but arises as semantic strengthening of the requirement $\psi \not\Rightarrow \phi$.

(35) $c + ye_{\psi}^{\text{EVEN}}(\phi)$ is defined only if

- a. $\psi \in \text{F-ALT}(\phi)$, focus-sensitivity
 - b. $c \subseteq \llbracket \psi \rrbracket$ and $\psi \not\Rightarrow \phi$ additivity
 - c. $\psi >_{c\text{-likely}} \phi$ (asymmetrically entailing $\psi \not\Rightarrow \phi$) scalarity
- when (a–c) are satisfied, $c + ye_{\vec{\psi}}^{\text{EVEN}}(\phi) = c + \phi$.

Finally, parallel to (15), the cc-potential definition of ‘even’ *ye* can be generalized to cases with multiple additive components, as schematized in (36). This generalized definition naturally captures the superlative use of ‘even’ (cf. Greenberg 2016, 2022): the prejacent proposition is less likely than all salient additive alternatives.

- (36) For any context c , sentence ϕ , and sequence of sentences $\vec{\psi}$:
- $$c + ye_{\vec{\psi}}^{\text{EVEN}}(\phi) = \begin{cases} c + \phi & \text{if } \forall \psi \in \vec{\psi} : c + \phi = c + \text{not-ONLY}_{\phi}^L(\psi) \\ \text{undefined} & \text{otherwise} \end{cases}$$

3.4. Remarks

This section examines how the proposed analysis accounts for the parallels and distinctions between ‘even’ *dou* and ‘even’ *ye*.

	logical strength	likelihood
<i>dou</i>	distributor	‘even’ (– additive)
<i>ye</i>	‘also’	‘even’ (+ additive)

Although *dou* and *ye* originate from different lexical functions—*dou* as a distributor and *ye* as an additive particle (‘also’)—both yield an ‘even’ interpretation when exclusivity is defined in terms of likelihood, a shift typically triggered in *lian*-constructions. In both cases, the asserted contribution of the particle is vacuous: for *dou*, the anti-exclusion inference collapses under the prejacent inference; for *ye*, there is no at-issue contribution to begin with. Nevertheless, both particles give rise to a scalar inference, which stems from either a lexically encoded presupposition or a definedness condition.

Specifically, for ‘even’ *dou*, the scalar inference comes from the non-vacuity presupposition, which requires the existence of at least one more likely alternative to the prejacent. For ‘even’ *ye*, by contrast, the scalar inference follows from the IAE requirement, which can be satisfied only when the additive component is contextually more likely than the asserted component.

Importantly, this shift to likelihood-based exclusivity affects the two particles differently. For *dou*, it results in semantic weakening: it does not require the existence of a logically weaker alternative but only a more likely one. Thus, *dou* unambiguously obtains an ‘even’ reading in contexts where the distributor reading is incompatible. For *ye*, however, the same shift leads to a semantic strengthening: wherever the additive ‘even’ reading is true, the ‘also’ reading is true as well. Consequently, ‘even’ *ye* relies more heavily on additional cues, such as the presence of *lian*, for disambiguation. This asymmetry explains the contrast observed in (30).

The two particles also differ with respect to additivity. ‘Even’ *ye* presupposes additivity because the exhaustifier encoded in IAE condition is the presuppositional exhaustifier ONLY_{ϕ}^L .

In contrast, ‘even’ *dou* is non-additive, since the exhaustifier in the anti-exclusion inference is the non-presuppositional exhaustifier O^L .

Beyond *lian*-sentences, the distribution of *dou* and *ye* across other scalar constructions is likewise shaped by their shared scalarity and their difference in additivity. The following sections discuss three such constructions. First, both particles can associate with minimizers, with largely parallel distributions (section 4). Second, both particles can function as an evaluative marker, though they trigger distinct evaluative inferences (section 5). Third, *ye* is more productive than *dou* in scalar concessives (section 6). I argue that only the minimizer-association use involves ‘even’ *ye*; in the latter two constructions, despite their scalarity character, *ye* functions as the non-scalar additive operator ‘also’.

4. Minimizer-association

4.1. Observations

In *lian ... dou/ye* constructions, the focused element can be a minimizer, which expresses the smallest quantity on a relevant scale. This use typically occurs with *dou/ye* is followed by negation, as illustrated in (37) and (38). This distribution parallels that of the minimizer-association use of English *even* (see Crnič, 2011, 2014), as reflected in the translations.

- (37) a. Mali (lian) YI-ge-ren dou/ye mei qing.
 Mary LIAN one-CL-person DOU/YE NEG invite
 ‘Mary didn’t even invite a SINGLe person.’
 b. Jintian (lian) YI-ge-ren dou/ye mei lai.
 today LIAN one-CL-person DOU/YE NEG come
 ‘Today not even a SINGLe person came.’
- (38) a. *Mali (lian) YI-ge-ren dou/ye qing-le.
 Mary LIAN one-CL-person DOU/YE invite-PERF
 ‘*Mary even invited a SINGLe person.’
 b. *Jintian (lian) YI-ge-ren dou/ye lai-le.
 today LIAN one-CL-person DOU/YE come-PERF
 ‘*Today even a SINGLe person came.’

Minimizers may also be licensed in the presence of an intensional expression like *xiang* ‘want’, as seen in (39).

- (39) Mali (lian) YI-fen-qian dou/ye xiang zheng.
 Mary LIAN one-cent-money DOU/YE want make
 ‘Mary wants to make any tiny amount of money.’

This configuration corresponds to what Crnič (2014) terms ‘weak *even*’, which arises in certain non-monotonic environments, as exemplified in (40a), but is subject to additional distributional constraints, as shown in (40b,c).

- (40) a. Exactly two congressmen read the constitution even ONCE.

- b. # Exactly four hundred congressmen read the constitution even ONCE.
- c. # Exactly two congressmen killed even ONE person.

Unlike English *even*, however, the minimizer associate in Mandarin *lian ... dou/ye* must be an internal argument (Guo, 1998; Xiang, 2020). More specifically, it may serve as the object of a transitive verb such as *qing* ‘invite’, as in (37a), or as the subject of an unaccusative predicate such as *lai* ‘come’, as in (37b). By contrast, it cannot serve as the subject of a transitive or unergative predicate, as shown in (41).

- (41) a. * (Lian) YI-ge-ren dou/ye mei qing Mali.
 LIAN one-CL-person DOU/YE NEG invite Mary
 Intended: ‘Not even a SINGle person invited Mary.’
- b. * (Lian) YI-ge-ren dou/ye mei xiao.
 LIAN one-CL-person DOU/YE NEG laugh
 Intended: ‘Not even a SINGle person laughed.’

The intended meaning in (41) can instead be expressed by modifying the head noun of the minimizer with a relative clause and having the entire minimizer phrase serve as the subject of the unaccusative predicate *you* ‘have/exist’.

- (42) a. (Lian) [YI ge qing-Mali-de ren] dou/ye mei you.
 LIAN one CL invite-Mali-MOD person DOU/YE NEG exist
 Lit. ‘There isn’t even ONE person who invited Mary.’
- b. (Lian) [YI ge xiao-de ren] dou/ye mei you.
 LIAN one CL laugh-MOD person DOU/YE NEG exist
 Lit. ‘There isn’t even ONE person who laughed.’

4.2. A reconstruction-based analysis

Building on Xiang’s (2020) account of *lian ... dou*, this section argues for a uniform reconstruction-based analysis for the licensing conditions of *lian-minimizer-dou/ye*. The scalarity of ‘even’ *dou/ye* requires that the asserted sentence is less likely than some alternative(s). When the focus is a minimizer, which expresses the smallest quantity on a scale, this requirement is satisfied only when this minimizer is semantically reconstructed to the c-commanding domain of a decreasing or non-monotonic operator. The LF of (37a) is thus represented as in (43).⁹

⁹This analysis also extends to cases like (i), where *dou/ye* associates with a predicate. ‘Try’ and ‘make a phone call’ are minimizers in the sense that they denote actions requiring minimal effort relative to alternatives such as ‘buy’ or ‘visit in person’.

- (i) a. Bieshuo mai le, Mali (lian) SHI dou/ye bu xiang shi.
 not.say buy LE Mary LIAN try DOU/YE NEG want try
 ‘Not to mention buying it, Mary didn’t even want to try it.’
- b. Bieshuo lai kan wo le, tamen (lian) DIANHUA dou/ye mei da-guo.
 not.say come see me LE, they LIAN call DOU/YE NEG make-EXP
 ‘Not to mention visiting me, they haven’t even given me a call.’

quantifiers, these NPIs are formed from *wh*-expressions and therefore have greater scope flexibility.

- (46) a. Mali SHEI ye/dou mei qing.
 Mary who YE/DOU NEG invite
 ‘Mary didn’t invite anyone.’
 b. SHEI ye/dou mei qing Mali.
 who YE/DOU NEG invite Mary
 ‘No one invited Mary.’

4.3. Interim summary

In sum, the similarities between ‘even’ *dou* and ‘even’ *ye* in *lian*-sentences extend to their largely identical distribution in minimizer-licensing sentences. The shared scalarity requirement forces the minimizer to be reconstructed into the c-commanding domain of a non-increasing operator. Once scalarity has been satisfied, additivity follows straightforwardly. The distributional constraint that the minimizer be an internal argument is reduced to general constraints on scope-taking with counting quantifiers.

5. Evaluative statements

5.1. Observations

Both *dou* and *ye* introduce an evaluative flavor when applied to scalar statements, but they give rise to different evaluative inferences. Specifically, *dou* implies that the prejacent is ranked relatively high on the relevant scale, as illustrated in (47); in contrast, evaluative *ye* implies that the prejacent is ranked modestly high, exceeding a minimal threshold, and it can be naturally paired with a concessive subordinate clause, as shown in (48).

- (47) a. Dou yijing WU-dian/ SHEN-ye le.
 DOU already five-o’clock/ dark-night LE
 ‘It’s already FIVE-o’clock/ late night.’
 ⇨ *It’s late.*
 b. Mali dou dadao DINGJI/ZHONGJI le.
 Mary DOU achieve elite.level/mid.level LE
 ‘Mary has already achieved the elite/mid-level.’
 ⇨ *Mary is competitive.*
- (48) a. (Jinguan tian hai mei hei,) ye yijing WU-dian le.
 Although sky still NEG dark YE already five-o’clock LE
 ‘(Although it hasn’t got dark yet,) It is already five-o’clock.’
 ⇨ *It’s late, even if not that late.*
 b. Mali (sui mei dadao dingji), ye dagao zhongji le.
 Mary although NEG achieve elite.level YE achieve mid.level LE
 ‘Mary has achieved the mid-level, (despite not yet the elite level).’
 ⇨ *Mary is more or less competitive, although not that competitive.*

With this contrast, *dou* is compatible with both extreme and non-extreme items, whereas *ye* can only associate with non-extreme items. When the prejacent clause is the highest-ranked alternative on the relevant scale, *ye* cannot receive an evaluative interpretation and instead defaults to a plain additive operator ('also'/'moreover'), as illustrated in (49).

- (49) a. Ye yijing SHEN-ye le.
 YE already dark-night LE
 'Moreover, it is already late night.'
 b. Mali ye dadao ZHONGJI le.
 Mary YE achieve mid.level LE
 'Moreover, Mary has already achieved the MID-level.'

Lian-sentences also allow for a scalar prejacent. A key feature distinguishing evaluative statements from *lian*-sentences is that evaluative statements obligatorily end with the evaluative marker *le*, as illustrated in (50). Crucially, this sentence-final *le* is not an aspectual marker: it can co-occur with the perfective suffix *-le*, as shown in (50b).

- (50) a. Mali lian JINPAI dou na -guo (le).
 Mary LIAN gold.medal DOU get -EXP LE
 'Mary has even got a gold medal.'
 b. Mali dou na -guo/le JINPAI *(le).
 Mary DOU get -EXP/PERF gold.medal LE
 'Mary has already got a gold medal.'

5.2. Explain evaluative *dou*

The evaluative use of *dou* is only briefly discussed in Xiang (2020). Under the anti-exhaustifier analysis, Xiang proposes that in such cases alternatives are ranked along a contextually determined scale, and sub-alternatives are those ranked lower than the prejacent on this scale. In this view, a sentence such as 'It is *dou* FIVE o'clock' can be paraphrased as 'It is five o'clock, not only four o'clock'. Given its scalar behavior, Xiang characterizes evaluative *dou* as a 'scalar marker'.

I formalize the semantics of evaluative *dou* as in (51), where scalarity is modeled using Wellwood's (2015) measure function (μ). In this formulation, *dou* applies to properties rather than propositions, since μ is only defined over objects that stand in a part-whole relation.

- (51) **Semantics of evaluative *dou***
 $\llbracket dou \rrbracket = \lambda f \lambda x \lambda w : \text{SUB}(f, C) \neq \emptyset. f(x)(w) = 1 \wedge \forall g \in \text{SUB}(f, C) [O_C^\mu(g)(x)(w) = 0]$
 a. $\text{SUB}(f, C) = \{g \mid g \in C \wedge \mu(g) < \mu(f)\}$
 b. $O_C^\mu = \lambda f \lambda x \lambda w. f(x)(w) = 1 \wedge \forall g \in C [g(x)(w) = 1 \rightarrow \mu(g) \leq \mu(f)]$

I propose that the evaluative flavor comes from the sentence-final particle *le*, which I define as an identity function with an evaluative presupposition, as in (52). For any property *f*, evaluative *le* presupposes that *f* lies within the domain of μ and that the degree to which *f* measures with respect to μ meets or exceeds the contextually determined standard associated with measure μ (viz., $s(\mu)$).

$$(52) \quad \llbracket le^{\text{EVAL}_\mu} \rrbracket = \lambda f : \text{EVAL}_\mu(f).f, \text{ where } \text{EVAL}_\mu(f) = 1 \text{ iff } f \in \text{Dom}(\mu) \wedge \mu(f) \geq \mathbf{s}(\mu).$$

Taken together, when evaluative *le* scopes over evaluative *dou*, the resulting evaluative inference is that the anti-exhaustified property meets or exceeds the relevant standard on the scale determined by μ . The examples in (47) are analyzed as follows:

- (53) ‘It is *dou* FIVE o’clock *le*.’ (= (47a))
- a. LF: [it [[*dou* [is FIVE o’clock]] *le*]] (*le* $\gg$ *dou*)
 - b. Assertion:
It is five o’clock, not just four.
 - c. Evaluative presupposition:
Being five and not just four o’clock is late.
- (54) ‘Mary *dou* achieved the MID-level *le*.’ (= (47b))
- a. LF: [Mary [[*dou* [achieved the MID-level]] *le*]] (*le* $\gg$ *dou*)
 - b. Assertion:
Mary has achieved the MID-level, not just a novice level.
 - c. Evaluative presupposition:
Achieved the MID-level and not just a novice level is competitive.

5.3. Explain evaluative *ye*

Turning to evaluative *ye*, the proposed schema for the semantics of ‘also’/‘even’ *ye* might initially suggest an analysis under which ‘It is *ye* FIVE o’clock’ is defined only if ‘It is five o’clock’ is contextually equivalent to ‘It is not only four o’clock (but rather five o’clock)’. While this idea is intuitively appealing, it encounters two immediate problems. First, under the assumption that *ye* acquires an additive presupposition by inheriting the prejacent presupposition of *only*, this analysis incorrectly predicts that ‘It is *ye* FIVE o’clock’ presupposes that it is four o’clock. Second, it fails to capture the distributional contrast between *dou* and *ye*, incorrectly predicting that *ye*, like *dou*, should be compatible with extreme items. In short, unlike in their ‘even’ uses in *lian*-sentences, evaluative *ye* in evaluative statements is not the additive counterpart of evaluative *dou*.

Instead, I propose that in an evaluative sentence, *ye*, unlike *dou*, takes scope above the evaluative marker *le*, as illustrated in (55a)/(56a). The IAE requirement of *ye* then applies not to the prejacent sentence itself, but to an evaluative inference concerning the status of the prejacent sentence on the relevant scale, as in (55b)/(56b), yielding a corresponding evaluative inference about an alternative, as in (55c)/(56c). This, in turn, derives the additive presupposition in (55d)/(56d).

- (55) [Context: speaking to someone planning to stay until late at night:]
‘It is *ye* FIVE o’clock *le*.’ (= (49a))
- a. LF: [it [*ye* [[is FIVE o’clock]] *le*]] (*ye* $\gg$ *le*)
 - b. Evaluative presupposition of *le*:
Five o’clock counts as (somewhat) late.

- c. IAE requirement of *ye*:
(b) is contextually equivalent to: *Not only does late night count as late.*
 - d. Additive presupposition inherited from (c):
Late night counts as late.
- (56) [Context: speaking to someone who regrets that Mary has not reached the elite level:]
‘Mary *ye* has achieved the MID-level *le*.’ (= (49b))
- a. LF: [Mary [[*le* [achieved the MID-level]] *ye*]] (*ye* $\gg$ *le*)
 - b. Evaluative presupposition of *le*:
The mid-level is (reasonably) competitive.
 - c. IAE requirement of *ye*:
(b) is contextually equivalent to: *Not only does the elite level count as competitive.*
 - d. Additive presupposition inherited from (c):
The elite level counts as competitive.

This account yields coherent predictions about the additive presupposition of *ye*. It also explains its incompatibility with extreme items and its concessive flavor: *ye* is felicitous only when the asserted component is ranked lower than the additive component on the relevant scale. A formal derivation is provided below.

The idea that evaluative *ye* targets evaluative inferences is further supported by its interaction between *dou*. As shown in (57), although *dou* alone cannot be paired with a concessive clause, it can be added after evaluative *ye*—provided that the prejacent corresponds to a mid-ranked item—without altering the overall meaning.

- (57) a. Jinguan tian hai mei hei, {*ye*, *ye-dou*, #*dou*} yijing WU-dian le.
Although sky still NEG dark YE YE-DOU DOU already five-o’clock LE
‘Although it is not dark yet, it’s already FIVE o’clock.’
- b. Suiran mei dao dingji, {*ye*, *ye-dou*, #*dou*} dao ZHONGJI le.
Although NEG achieve elite.level, YE YE-DOU DOU achieve mid.level LE
‘Although not yet achieving the elite level, (she) has achieved the mid-level.’

Under the proposed analysis, the *ye-dou* clauses in (57a,b) have the LFs in (58a,b), respectively, where *le* takes scope below *ye* but above *dou*. The underlined portions in the paraphrases reflect the anti-exclusion inference contributed by *dou*: *dou* anti-excludes a lower-ranked scalar alternative and thus forms part of the asserted component to which *le* and *ye* apply.

- (58) a. LF: [... [*ye* [[*dou* [is FIVE o’clock]] *le*]]] (*ye* $\gg$ *le* $\gg$ *dou*)
‘... [five and not just four] o’clock also counts as late.’
- b. LF: [... [*ye* [[*dou* [achieved the MID-level]] *le*]]] (*ye* $\gg$ *le* $\gg$ *dou*)
‘... [achieving the mid-level and not just the novice-level] is also competitive.’

A cc-potential definition of an evaluative *ye*-sentence is given in (59). As argued, *ye* and *le* have no at-issue contribution, but they each introduce a definedness condition, namely, (i) the evaluative presupposition of *le*, and (ii) the IAE requirement of *ye*.¹²

¹²The two definedness conditions can also be represented with a single equation:

(59) **Semantics of evaluative *ye***

For any context c , measure function μ , properties f and g , and individual a :

$$c + ye_{\text{EVAL}_\mu(g)}(le^{\text{EVAL}_\mu(f)})(a) = \begin{cases} c + f(a) & \text{if (i) } c \subseteq \llbracket \text{EVAL}_\mu(f) \rrbracket \\ & \text{(ii) } c + \text{EVAL}_\mu(f) = c + \text{not-ONLY}_{\text{EVAL}_\mu(f)}(\text{EVAL}_\mu(g)) \\ \text{undefined} & \text{otherwise} \end{cases}$$

Notably, despite the scalar character of evaluative *ye*-sentences—arising from the evaluative inferences about the status of the asserted and the additive components on the scale—evaluative *ye* is not itself scalar. Unlike the scalar O^μ -operator defined in (51), the exhaustifier encoded in the IAE requirement of evaluative *ye* follows the logical-strength-based definition in (14a), under which exclusion targets alternatives not entailed by the prejacent. In this respect, it is more accurate to say that evaluative *ye* applies to scalar sentences, rather than that evaluative *ye* itself has a scalar interpretation.

The cc-potential definition in (59) is semantically equivalent to (60). Crucially, in (60d), the condition $\text{EVAL}_\mu(g) \not\Rightarrow \text{EVAL}_\mu(f)$ follows from the IAE requirement. It states that g satisfying the standard associated with μ does not entail f satisfying that standard. This condition is satisfied only when f is ranked lower than g ; if f and g occupy the same rank, we would instead have $\text{EVAL}_\mu(g) \Leftrightarrow \text{EVAL}_\mu(f)$. Consequently, the additive presupposition induced by the IAE requirement results in a concessive flavor.

- (60) $c + ye_{\text{EVAL}_\mu(g)}(le^{\text{EVAL}_\mu(f)})(a)$ is defined only if
- a. $c \subseteq \llbracket \text{EVAL}_\mu(f) \rrbracket$ evaluativity
 - b. $g \in \text{F-ALT}(f)$ focus-sensitivity
 - c. $f, g \in \text{Dom}(\mu)$ scalarity
 - d. $c \subseteq \llbracket \text{EVAL}_\mu(g) \rrbracket$ and $\text{EVAL}_\mu(g) \not\Rightarrow \text{EVAL}_\mu(f)$ additivity
or equivalently, $c \subseteq \llbracket \text{EVAL}_\mu(g) \rrbracket$ and $\mu(f) < \mu(g)$ concessive flavor
- when (a–d) are satisfied, $c + ye_g^{\text{EVAL}_\mu(f)}(a) = c + f(a)$.

5.4. Interim summary

In sum, *dou* and *ye* give rise to different evaluative inferences in evaluative statements. This section attributes these differences to their relative scope with respect to the evaluative marker *le*. Specifically, *dou* scopes below *le*, functioning as an anti-exhaustifier that anti-excludes lower-ranked alternatives. By contrast, *ye* scopes above *le*, and its IAE requirement applies to the evaluative presupposition of *le*, rather than to the literal prejacent.

The wide-scope interpretation of *ye* is driven by its additivity: the additive presupposition cannot be satisfied if *ye* takes scope below *le*, but it can be satisfied when *ye* scopes above *le*, provided that the asserted component is ranked lower than a salient additive alternative.

Finally, this section argues that the evaluative use of *ye* is independent of the scalarity effect associated with its ‘even’ use. While evaluative *dou* has a genuinely scalar semantics,

(i) $c = c + \text{EVAL}_\mu(f) = c + \text{not-ONLY}_{\text{EVAL}_\mu(f)}(\text{EVAL}_\mu(g))$

defined in terms of scale-based exclusivity, evaluative *ye* functions as a non-scalar additive operator ‘also’.

6. Scalar concessives

6.1. Observations

A *ye*-clause frequently co-occurs with a concessive subordinate clause, yielding an ‘even if’ or ‘although’/‘even though’ interpretation. These two interpretations differ in their factual status: ‘even if’ introduces a non-factual or hypothetical antecedent, whereas ‘although’ presupposes the truth of the subordinate clause. Illustrative examples are given in (61) and (62), where underlining marks the concessive subordinators.

(61) ‘Even if’

- a. Mali jishi mang ye/?dou hui chou-kong duanlian.
Mary JISHI busy YE/DOU will make-time workout
‘Mary will make time to do workout even if she is busy.’
- b. Jiusuan zhe-ben-shu youdian-gui, wo ye/?dou yao mai.
JIUSUAN this-CL-book a.little-expensive, I YE/DOU will buy
‘Even if this book is a little expensive, I would (still) buy it.’

(62) ‘Although’/‘Even though’

- a. Mali jinguan mang ye/*dou jingchang chou-kong duanlian.
Mary JINGUAN busy YE/DOU often make-time workout
‘Mary often makes time to do workout despite that she is busy.’
- b. Jinguan zhe-ben-shu youdian-gui, wo ye/*dou mai-le.
JINGUAN this-CL-book a.little-expensive, I YE/DOU buy-PERF
‘Although this book is a little expensive, I (still) bought it.’

I call both types of constructions ‘scalar concessives’. In particular, sentences with an ‘even if’ interpretation are called ‘scalar concessive conditionals’ (Haspelmath & König, 1998), and sentences with an ‘although’ interpretation are called ‘genuine scalar concessives’. In Mandarin, both types of scalar concessives are formed by combining a concessive subordinator with *ye*, differing only with respect to the choice of the subordinator. Possible realizations are summarized in (63), where S and S’ stand for the main clause and the subordinate clause, respectively.

- (63) a. Forms of scalar concessive conditionals ($\approx$ ‘even if’):
jishi/jiusuan/jibian/napa S’, ye S.
- b. Forms of genuine scalar concessives ($\approx$ ‘although’):
jinguan/suiran S’, ye S. (Or equivalently: *S, jinguan/suiran S’.*)

The examples above imply that the condition expressed in the subordinate clause counts against the truth of the main clause. However, as illustrated in (64), concessives can also be

used in cases where the subordinate clause does not count against the truth of the main clause itself, but rather against that of a salient alternative that is ranked higher than the main clause or as high as the main clause. For instance, in (64a), making one mistake in an exam would prevent Mary from getting an A but still grant her a B. In (64d), that Mary not helping us renders it false that Mary will help us, an alternative ranked comparable to the main clause ‘John will help us’. I henceforth refer to the cases in (61) and (62) ‘result-preserving’ and those in (64) as ‘result-changing’.

- (64) a. Jibian/suiran ta da-cuo-le yi-dao-ti, ye/*dou hui de B.
 JIBIAN/JIUSUAN 3SG answer-wrong-perf one-CL-question, YE/DOU will get B.
 ‘Even if/though she made one mistake, she would get a B.’
- b. Ta jishi bu lai, ye/*dou yinggai da-ge-dianhua.
 3SG JISHI NEG come, YE/DOU should make-CL-call
 ‘Even if she is not coming, she should call us.’
- c. Jinguan ta ziji mei lai, ye/*dou pai-le daibiao lai le.
 JINGUAN 3SG self NEG come, YE/DOU send-perf delegate come LE
 ‘Although she could not come, she sent a delegate to come.’
- d. Jiusuan Mali bu bang women, Yuehan ye/*dou hui bang women.
 JIUSUAN Mary NEG help we, John YE/DOU will help we.
 ‘Even if Mary doesn’t help us, John will help us.’

Interestingly, despite exhibiting a similar ‘even’ interpretation in *lian*-sentences, *dou* is not productively used in concessive sentences: in all the scalar concessive examples above, replacing *ye* with *dou* results in deviance or even ungrammaticality. However, *dou* is acceptable when paired with a concessive clause denoting an extreme condition that counts against the truth of the main clause, as exemplified in (65), where it highlights the extreme unlikelihood of someone persisting in the face of a risk of bankruptcy.

- (65) Jibian/jinguan you pochan-de-fengxian, wo ye/dou bu-hui fangqi.
 JIBIAN/JINGUAN exist bankrupt-MOD-risk, I YE/DOU NEG-will give.up
 ‘I won’t give up even if/ despite that there is a risk of bankruptcy.’

6.2. Why ‘even if’ analyses do not extend to Mandarin

There is rich literature on the semantics of *even if* sentences in English. A broad consensus holds that the meaning of an *even if* sentence is compositionally derived by applying the scalar particle *even* to an indicative *if*-conditional, with the concessive flavor arising from the scalar effect contributed by *even* (König, 1985, 1991). This is also a generalization reached by Forker’s (2016) typological study on additive expressions: in all languages examined in this study, additive expressions that develop a concessive use all have a scalar interpretation.

However, this line of analysis does not extend straightforwardly to Mandarin. First, the particle *dou*, despite admitting a scalar interpretation, has a relatively restricted distribution in concessive constructions, as seen in section 6.1. Second, unlike in English, scalar concessives

in Mandarin cannot be formed based on the regular antecedent connective *ruguo* 'if'. For example, (66) only admits a simple additive reading; it presupposes that there is someone other than Mary that we would reserve a seat for if they come, but not that Mary is a less likely candidate to get a seat reservation.

- (66) Ruguo MALI yao lai, we ye hui gei ta liu weizi.
if Mary will come, I YE will for 3SG reserve seat
'If Mary_F is coming, I will reserve a seat for her as well.'

To further clarify this issue, it is useful to see where Mandarin sits in the typology of scalar concessives. Haspelmath & König (1998) identify two main strategies for forming scalar concessive conditionals in European languages. The more common strategy involves an ‘even/also + if’ construction, in which an additive expression combines with an ‘if’-clause. Importantly, in contrast to Forker’s (2016) generalization, the additive expression may be scalar (e.g., English *even*) or non-scalar (e.g., Albanian *edhe*, Italian *anche*, Finnish *-kin*, German *auch*, and Hebrew *gam*; all glossed as ‘also’). The alternative strategy employs a dedicated concessive subordinator, the same as forming a genuine concessive (e.g., Dutch *al*, Finnish *vaikka*, Lithuanian *tegu*, Spanish *aunque*, and Upper Sorbian *byrnjež*; all glossed as ‘though’).

Compared to these two strategies, Mandarin scalar concessives, including both concessive conditionals and genuine concessives, exhibit a hybrid form that combines a subordinator with an additive expression (e.g., *ye* 'also', *haishi/rengran* 'still'). As shown in (61) and (62), omitting *ye* or replacing it with the non-additive particle *dou* results in deviance or ungrammaticality.

Observations in section 6.1 and the cross-linguistic comparison in section 6.2 raise two central questions: (i) how are concessivity and scalarity realized in Mandarin scalar concessives, and (ii) why, in contrast to European languages, Mandarin scalar concessives usually have the hybrid form ‘CONC + *ye*’, and occasionally, ‘CONC + *dou*’. Sections 6.3 and 6.4 address these two questions, considering both result-preserving and result-changing cases.

6.3. Deriving concessivity and scalarity

6.3.1. Composing result-preserving scalar concessives

For result-preserving concessives, I assume that the concessive subordinator (CONC) presupposes that the subordinate clause counts against the truth of the main clause under stereotypical expectations (Müller, 2016). The particle *ye* contributes to an additive presupposition, which is realized as a normality inference that the main clause being true under the salient alternative condition confronts to normality. These two conditions are henceforth referred to as ‘Abnormality’ and ‘Alternative Normality’, respectively.

- (67) 'CONC α , $\text{ye } \beta$ ' (with the result-preserving alternative: *if* α' , β)
- a. Normally, if α , not β . (Abnormality; presupposed by CONC)
 - b. Normally, if α' , β . (Alternative Normality; presupposed by ye)

The Abnormality presupposition is the source of **concessivity** (Müller, 2016): the truth of $[if\ \alpha, \beta]$ in the actual world, despite its unlikelihood, entails that the actual world is not among the best worlds with respect to conformity to normality.

Furthermore, the Abnormality condition, together with the Alternative Normality condition from the additivity of *ye*, yields a **scalarity** requirement according to which the alternative conditional $[if\ \alpha', \beta]$ is more likely than $[if\ \alpha, \beta]$. As a result, the ‘also’–‘even’ distinction of the additive particle *ye* effectively collapses, since scalarity has been independently contributed by other presuppositional properties.

This prediction aligns with Faller’s (2022) account of the ‘also + if’ type concessive conditionals in German. Building on Müller’s analysis of concessivity in genuine concessives, Faller argues that it is the conditional perfection reading of the additive presupposition that yields an abnormality condition, which in turn gives rise to concessivity.

Composing genuine scalar concessives For a genuine scalar concessive, I assume the LF in (68a) and the structure of the additive component in (68b).

- (68) ‘*Jinguan* this book was expensive, I *ye* bought it.’
- a. $[ye_\psi [\phi\ jinguan^\square (f[\alpha\ \text{this book was expensive}]) (o) [\beta\ \text{I bought it}]]]$
 - b. where ψ : $[if^\square (f[\alpha'\ \text{this book were inexpensive}]) (o) [\beta\ \text{I bought it}]]]$

The asserted component asserts the implication (69a) and presupposes the truth of the antecedent as in (69b). Moreover, it presupposes the unlikelihood of buying this book when it is expensive, modeled in terms of Kratzer’s (1981) stereotypical ordering source as in (69c). Taken together, the three inferences yield concessivity: they are simultaneously true in w only when w is not a best-ranked world relative to the stereotypical ordering source.

- (69) Inferences from the asserted component ϕ :
- $[\phi\ jinguan^\square (f[\alpha\ \text{this book was expensive}]) (o) [\beta\ \text{I bought it}]]]$
 - a. $w \in \llbracket \alpha \rrbracket \rightarrow w \in \llbracket \beta \rrbracket$ (Assertion)
(If it’s true that this book was expensive, then it’s true that I bought this book.)
 - b. $w \in \llbracket \alpha \rrbracket$ (Factivity)
(It is true that this book was expensive.)
 - c. $\forall v \in \text{Best}_{o(w)}(f(w) \cap \llbracket \alpha \rrbracket) : v \notin \llbracket \beta \rrbracket$ (Abnormality)
(Among the worlds that are similar to the utterance world and in which the book is expensive, the best-ranked ones relative to the stereotypical ordering source (o) are worlds in which I did not buy this book.)

The additive component ψ is an indicative conditional, interpreted as ‘Normally, if α' then β' ’, as in (70a). Abnormality (69c) and Alternative Normality (70a) together entail a scalarity inference that buying this book when it is inexpensive is more likely than buying this book when it is expensive. I also assume that an unmarked ‘if’-conditional implies that the utterance world is best-ranked among the worlds where the antecedent is true, as in (70b).

- (70) Inferences from the additive component ψ :
- $[\psi\ if^\square (f[\alpha'\ \text{this book were inexpensive}]) (o) [\beta\ \text{I bought it}]]]$
 - a. $\forall v \in \text{Best}_{o(w)}(f(w) \cap \llbracket \alpha' \rrbracket) : v \in \llbracket \beta \rrbracket$ (Alternative Normality)
(Among the worlds that are similar to the utterance world and in which the book

is inexpensive, the best-ranked ones relative to the stereotypical ordering source (o) are worlds in which I bought this book.)

- b. $w \in \text{Best}_{o(w)}(f(w) \cap \llbracket \alpha' \rrbracket)$

Composing scalar concessive conditionals

Scalar concessive conditionals are somewhat more complex because the consequent is modalized. My analysis builds on the nested-modal theory of conditionals (Frank, 1996; von Stechow & Iatridou, 2005). This theory adopts the restrictor analysis of Kratzer (1981, 1986, 1991) that an indicative antecedent clause restricts the domain of a modal expression, but it assumes that the modal being restricted is an epistemic necessity modal globally applied to the entire conditional, not the modal in the consequent clause. This nested-modal analysis straightforwardly accounts for the normality and abnormality inferences.

I assume the LF in (71a) and the structure of a result-preserving alternative as in (71b). The epistemic necessity modal encoded within the subordinator *jishi* associates with an epistemic modal flavor (f_1) and a stereotypical ordering source (o_1), whereas the modal auxiliary *would* in the consequent clause associates with a circumstantial modal flavor (f_2) and a bouletic ordering source (o_2). Other assumptions parallel the case of the genuine scalar concessive. For readability, modal flavor is omitted in the paraphrases.

- (71) ‘*Jishi* this book is expensive, I *ye* would buy it.’

- a. $[ye_\psi [\phi \text{ jishi}^\square (f_1 [\alpha \text{ this book is expensive}]) (o_1) [\text{would}^\square f_2 (o_2) [\beta \text{ I buy it}]]]]$
b. $\psi: [\text{if}^\square (f_1 [\alpha' \text{ this book is inexpensive}]) (o_1) [\text{would}^\square f_2 (o_2) [\beta \text{ I buy it}]]]$

- (72) Inferences from the asserted component ϕ :

$[\phi \text{ jishi}^\square (f_1 [\alpha \text{ this book is expensive}]) (o_1) [\text{would}^\square f_2 (o_2) [\beta \text{ I buy it}]]]$

- a. Assertion

$w \in \llbracket \alpha \rrbracket \rightarrow [\forall u \in \text{Best}_{o_2(w)}(f_2(w)) : u \in \llbracket \beta \rrbracket]$

(If it is true that this book is expensive, it is true that I would buy this book.)

- b. Abnormality

$\forall v \in \text{Best}_{o_1(w)}(f_1(w) \cap \llbracket \alpha \rrbracket) : \neg \forall u \in \text{Best}_{o_2(v)}(f_2(v)) : u \in \llbracket \beta \rrbracket$

(In all of the best worlds in which this book is expensive, ranked relative to the stereotypical ordering source (o_1), it is false that my buying this book necessarily best complies my desires (o_2).)

- (73) Inferences from the additive component ψ :

$[\psi \text{ if}^\square (f_1 [\alpha' \text{ this book is inexpensive}]) (o_1) [\text{would}^\square f_2 (o_2) [\beta \text{ I buy it}]]]$

- a. Alternative Normality

$\forall v \in \text{Best}_{o_1(w)}(f_1(w) \cap \llbracket \alpha' \rrbracket) : \forall u \in \text{Best}_{o_2(v)}(f_2(v)) : u \in \llbracket \beta \rrbracket$

(In all of the best worlds in which this book is inexpensive, ranked relative to the stereotypical ordering source (o_1), it is true that my buying this book necessarily best complies my desires (o_2).)

- b. $w \in \text{Best}_{o_1(w)}(f_1(w) \cap \llbracket \alpha' \rrbracket)$

6.3.2. Extending to result-changing scalar concessives

In result-changing cases, the concessive subordinator and the additive particle *ye* also trigger Abnormality and Alternative Normality, respectively, as illustrated in (74). However, unlike in result-preserving cases, these conditions do not target the asserted main clause (β), but rather an equally-ranked or higher-ranked alternative (β^+). The ranking is based on a contextually determined ordering source.

- (74) [CONC α , *ye* β] (with the result-changing alternative: [*if* α' , β^+])
- a. Normally, if α , not β^+ . (Abnormality; presupposed by CONC)
 - b. Normally, if α' , β^+ . (Alternative Normality; presupposed by *ye*)

For instance, sentence (75) states that missing one question would prevent Mary from getting an A but would still grant her a B, where ‘getting an A’ is ranked higher than ‘getting a B’ with respect to Mary’s goals or desires. Notably, while in a result-preserving case the assertion is a simple implication (as seen in (69a) and (72a)), here the assertion can be read as an indicative conditional: ‘Normally, if Mary missed one question, she would get a B’.

- (75) *Jinguan* Mary missed one question (α), she *ye* would get a B (β).
 $\approx$ *If Mary did perfect, she would get an A. If she missed one question, she would get a B.*
- a. Abnormality:
If Mary missed one question (α), she wouldn’t get an A (not β^+).
 - b. Alternative normality:
If Mary did perfect (α'), she would get an A (β^+).

More examples are given below:

- (76) *Jishi* she does not have time to come (α), she *ye* should call us (β).
- a. If she does not have time to come (α), she doesn’t have to come (not β^+).
 - b. If she has time to come (α'), she should come (β^+).
(β^+ is ranked higher than β with respect to conventional social obligations.)
- (77) *Jishi* Mary cannot help us (α), John *ye* will help us (β).
- a. If Mary cannot help us (α), she won’t help us (not β^+).
 - b. If Mary can help us (α'), she will help us (β^+).
(β^+ and β are equally ranked with respect to our goals.)

I now turn to the question of how concessivity and scalarity are realized in result-changing concessives. First, since Abnormality does not bear directly on the asserted conditional, it does not contrast with the truth of the asserted conditional in the actual world, and therefore does not preclude the actual world from being among the best-ranked worlds with respect to conformity to normality. Instead, concessivity arises from the rank of the main clause, namely, that the condition given by the subordinate clause renders only the truth of the main clause, but not that of a higher or equally ranked alternative. Second, while Abnormality and Alternative Normality derive a scalar condition, wherein [*if* α , β^+] is less likely than [*if* α' , β^+], this condition does not affect the likelihood of the asserted conditional.

6.4. Distributing *ye* and *dou* in scalar concessives

6.4.1. Why ‘CONC + *ye*’?

Having explained the source of concessivity and scalarity in scalar concessives, I now turn to the question of why Mandarin scalar concessives exhibit a hybrid form, combining a concessive subordinator with an additive expression (*ye* ‘also’ or *haishi/rengran* ‘still’).

When viewed in the broader context of subordinate constructions, this hybrid form is not unexpected. In Mandarin, subordinate clauses are typically paired with a corresponding connective or adverb in the main clause that yields a parallel discourse effect. Some illustrative examples are given in (78). By contrast, in the English translations, such corresponding elements are often omitted because they sound redundant, as seen in (78a,b), or they lack a direct counterpart, as seen in (78c,d).

- (78) a. Zhe-ben-shu suiran hen jiu, *(dan) neirong hen hao.
 this-CL-book although very old, but content very good.
 ‘Although this book is very old, but its content is very good.’
 b. Yinwei zuotian shui-wan-le, *(suoyi) jintian qi-wan-le.
 because yesterday sleep-late-PERF, therefore today get.up-late-PERF
 ‘Because I went to bed late last night, therefore I got up late today.’
 c. Ruguo ni bu lai, wo *(jiu) hui-jia qu.
 if you NEG go, I IJU return-home go.
 ‘If you are not coming, I will go home.’
 d. Zhiyou da-dui zhe-dao-ti, ta *(cai) neng jige.
 only answer-correct this-CL-question, 3SG CAI can pass
 ‘Only if she answers this question correctly, she can pass.’

I argue that CONC pairs with *ye* because they share closely related discourse functions. The relation between CONC and *ye* in a concessive is analogous to that between *because* and *therefore*: both pairs encode complementary perspectives on the same underlying relation. If *p* is a reason for *q*, then *q* is a consequence of *p*, and vice versa. Similarly, in scalar concessives, the Abnormality presupposition contributed by CONC and the Alternative Normality presupposition contributed by *ye* can be derived from one another via conditional perfection-type strengthening (cf. Faller 2022).

- (79) ‘CONC α , *ye* β ’ (with alternative: if not α , β)
 a. Normally, [if α , not β]. (Abnormality; presupposed by CONC)
 b. Normally, [if not α , β]. (Alternative Normality; presupposed by *ye*)

This proposal is further supported by the observation that both concessives and additives exhibit a relevance- or cause-canceling effect. The sentences in (80) are infelicitous due to default relevance or causal inferences: (80a) suggests that the speaker is more likely to buy a book if it is expensive, and (80b) suggests that the book’s high price caused the purchase. Such effects have been analyzed in terms of conventional meaning, conversational implicatures, or principles of discourse coherence (see Lassiter, 2023, and references therein).

- (80) a. # If this book is expensive, I would buy it.
 b. # This book was expensive, I bought it.

Crucially, these effects disappear when the sentence is marked as concessive, as seen in (81), or when an additive particle (e.g., *also*, *still*) appears in the main clause, as seen in (82) (König & Siemund, 2000; Kehler, 2002; Lund, 2017; Lassiter, 2023).

- (81) a. Even if this book is expensive, I would buy it.
 b. Although this book was expensive, I bought it.
 (82) a. Not only I would buy this book if it inexpensive—if it is expensive, I would also buy it.
 b. This book was expensive, and I still bought it.

This analysis makes promising predictions for the cross-linguistic typology of scalar concessives. Recall that in European languages scalar concessives may take the form of ‘even + if’ or ‘also + if’, or employ a dedicated concessive subordinator (Haspelmath & König, 1998). While earlier work on ‘even + if’ type concessives derives concessivity from the scalarity effect of ‘even’, the proposed analysis explains why scalar concessives may be formed without a scalar operator. In contexts involving an epistemic necessity modal, the additive particle and the concessive subordinator can each give rise to the function of the other. Specifically, when an additive expression is used, the additive presupposition is realized as an Alternative Normality condition, which, upon strengthening, yields Abnormality (Faller, 2022). Conversely, in concessives formed with a single concessive subordinator, the subordinator presupposes Abnormality, which, upon strengthening, gives rise to Alternative Normality.

6.4.2. Why occasionally ‘CONC + *dou*’?

The particle *dou* is not an ideal matching word for concessive subordinators. As a non-additive expression, ‘even’ *dou* presupposes the existence of some more likely alternatives but does not require them to be true, which is too weak to derive Abnormality.

Then, why is *dou* acceptable in concessives like (83) (repeated from (65))? This sentence emphasizes that the consequent holds under any condition, including a condition that strongly counts against it.

- (83) Jibian/jinguan you pochan-de-fengxian, wo ye/dou bu-hui fangqi.
 JIBIAN/JINGUAN exist bankrupt-MOD-risk, I YE/DOU NEG-will give.up
 ‘I won’t give up even if/ despite that there is a risk of bankruptcy.’

I argue that here *dou* is not paired with the concessive subordinate clause, but rather a covert unconditional *wulun*-clause. The intended structure can be spelled-out as in (84). Here the concessive clause introduces an extreme condition, with the effect of widening the domain of the unconditional clause:

- (84) Wulun you shenme fengxian, jibian you pochan-de-fengxian, wo ye/dou
 no.matter exist what risk, JIBIAN exist bankrupt-MOD-risk, I YE/DOU
 bu-hui fangqi.
 NEG-will give.up

‘Whatever risk there is, even if there is a risk of bankruptcy, I won’t give up.’

As predicted, a scalar concessive cannot take the form ‘CONC + *dou*’ if the main clause is incompatible with an unconditional. This incompatibility is language independent, as illustrated in English below:

- (85) a. ? Whatever the price, even if it is a bit expensive, I would buy it. (cf. (61b))
(Slightly deviant, because the concessive condition is redundant.)
- b. * Whatever the price, although it was a bit expensive, I bought it. (cf. (62b))
(Ungrammatical, because ‘whatever’ is incompatible with an episodic consequent.)
- c. # No matter how Mary performed perfect in the exam, even if she made one mistake, she would get a B. (cf. (64a))
(Infelicitous, because it entails a false inference that Mary would get a B when she did perfect in the exam.)
- d. # No matter whether she has time to come, even if she does not have time to come, she should call us. (cf. (64a))
(Infelicitous, because it entails a false inference that Mary should call us when she has time to come (and will actually come).)

It is worth noting that when *dou* is associated with an unconditional clause, its function is not ‘even’, but rather a FC-trigger. Under Xiang’s (2020) anti-exhaustifier analysis, *dou* obtains a FC-trigger function when anti-exclusion targets not the more likely alternatives but the so-called ‘non-innocently-excludable alternatives’, such as the disjuncts of a disjunction (Fox, 2007). In this use, *dou* forces a disjunctive sentence to receive a FC interpretation, as seen in (86a), and it can freely associate with FCIs (such as *wh/renhe*-phrases or unconditional clauses) in FC-licensing contexts, as exemplified in (86b,c).

- (86) a. [Yuehan huozhe Mali] (*dou/ye*) keyi jiao jichu-hanyu.
John or Mary *DOU/YE* can teach intro-Chinese
Without *dou/ye*: ‘Either John or Mary can teach Intro Chinese.’
With *dou*: ‘Both John and Mary can teach Intro Chinese.’
With *ye*: ‘John or Mary also can teach Intro Chinese.’
- b. [Shei/shenme-ren/renhe-ren] *dou/?ye* you-jihui tongguo ceshi.
who/what-person/any-person *DOU/YE* have-chance pass test
Intended: ‘Anyone/everyone has a chance to pass the test.’
- c. [Wulun baitian haishi wanshang], ta *dou/*ye* yao kai-deng.
no.matter day or night, 3SG *DOU/YE* want open-light
‘She keeps lights on no matter whether it is day time or night time.’

By contrast, *ye* cannot trigger FC: in (86a), associating *ye* with a disjunction yields only its regular ‘also’ interpretation. Moreover, when associated with polarity items, *ye* is subject to significant distributional restrictions (Yang, 2019; Liu, 2023; Xiang, 2025). In (86b,c), replacing *dou* with *ye* results in deviance or ungrammaticality.

6.5. Interim summary

Mandarin scalar concessives take a hybrid form, ‘CONC α , *ye* β ’, combining a concessive subordinate clause with a main clause that contains an additive particle. The subordinator CONC presupposes Abnormality. In result-preserving cases, this condition is read as ‘Normally, if α , not β ’. Given the asserted implication, it follows that the actual world is not among the best worlds ranked by conformity to normality, thereby yielding a concessive flavor (Müller, 2016). In result-changing cases, the Abnormality condition is read as ‘Normally, if α , not β^+ ’, where β^+ is ranked as least as high as β with respect to a contextually determined ordering source. In such cases, concessivity arises from the non-superior ranking of β relative to β^+ .

Furthermore, when the additive component takes the form of an indicative conditional, the additive presupposition of *ye* is realized as an Alternative Normality condition: ‘Normally, if α' , not $\beta^{(+)}$ ’. When combined with Abnormality, this yields a scalar inference that [if α , then $\beta^{(+)}$] is less likely than [if α' , then $\beta^{(+)}$].

Moreover, since Abnormality and Alternative Normality can be derived from each other through conditional perfection-type strengthening (Faller, 2022), a sentence may obtain concessivity as long as its literal meaning entails either condition (or both). This prediction naturally accounts for the cross-linguistic variation in the formation of scalar concessives.

Finally, this section has argued that neither *ye* nor *dou* functions as ‘even’ in scalar concessives. In ‘CONC + *ye*’ constructions, scalarity follows from the interaction between Abnormality and Alternative Normality. In ‘CONC + *dou*’ constructions, which are acceptable only in limited cases, *dou* instead functions as a FC-trigger, paired with a covert unconditional clause.

7. ‘Independence’ and Fluctuation

7.1. A remaining issue on Independence and a QUD-based account

Both ‘also’ and ‘even’ are subject to an independence requirement that the asserted prejacent ϕ_A neither entails nor is entailed by the additive component ψ_P (A and P stand for assertion and presupposition, respectively) (Jasinskaja & Zeevat, 2009; Beaver & Clark, 2008; Theiler, 2019, among others). Some English examples are given below. The same pattern applies to Mandarin *ye* (for ‘also’ cases, see Liu 2023; for ‘even’ cases, see example (20)).

(87) Violations of $\psi_P \not\Rightarrow \phi_A$:

- a. John and Mary came, # John_F also came.
- b. Not to mention John and Mary, # even John_F came.
- c. Everyone came, # John_F also came.
- d. Not to mention EVERYone, # even John_F came.

(88) Violations of $\psi_P \not\Leftarrow \phi_A$:

- a. John came, # [John and Mary]_F also came.
- b. Not to mention John, # even [John and Mary]_F came.
- c. John came, * EVERYone_F also came.

- d. Not to mention John, * even EVERYone_F came.

Previous sections have derived the requirement of $\psi_P \not\Rightarrow \phi_A$ from IAE. Then, what about the other direction of independence, namely, $\psi_P \neq \phi_A$?

I argue that the infelicity of the illustrative cases in (88) is due to a pragmatic constraint that presupposed content cannot be at-issue with respect to the question-under-discussion (QUD). When the additive and asserted components are mutually independent, the *also*-sentence may address a well-formed ‘*wh*-else’-question that excludes the additive alternative from the QUD, as shown in (89). In contrast, the ‘*also*’-sentence (90) is infelicitous because there is no felicitous QUD to which it may serve as a possible answer.

- (89) John came, Mary_F also came.
QUD: ‘Knowing that John came, who else came?’
- (90) John came, # [John and Mary]_F also came.
a. QUD 1: Knowing that John came, who else came?
Infelicitous: ‘John and Mary came’ isn’t a possible answer to this QUD.
b. QUD 2: # Knowing that John came, who came?
Infelicitous: the QUD itself is infelicitous.

This analysis also extends to ‘*even*’-sentences, regardless of additivity, as exemplified in (91).

- (91) Even MArY came. (Alternative: John came)
a. QUD for additive ‘*even*’: ‘Knowing that John came, who else came?’
b. QUD for non-additive ‘*even*’: ‘Putting whether John came aside, who else came?’

The QUD-based analysis is supported by three observations. First, as exemplified in (92), $\psi_P \Leftarrow \phi_A$ is permitted when the presupposed additive component ψ_P is only entailed by the presupposition of ϕ_A , not by the assertion of ϕ_A .

- (92) a. Mary participated in the game. She also/even WON_F the game.
(QUD: Knowing that Mary participated in the game, what did she achieve in the game?)
b. John believes that Mary came. # He also/even KNOWs_F it.
(QUD: # Knowing that John believes that Mary came, does he KNOW it?)

Second, as a result of a general pragmatic principle, the infelicity of (90) is independent of the presence of *also*. Below, (93a) is infelicitous because the presupposed inference ‘John came’ is at-issue w.r.t. the QUD ‘Who came?’. In contrast, in (93b), adding emphatic *BOTH* improves the sentence despite not changing the truth-conditional meaning: now the QUD takes ‘only John came’, as opposed to ‘John came’, as at-issue content.

- (93) a. # Not only JOHN_F came, [John and MArY]_F came.
(Because ‘John came’ is at-issue w.r.t. the QUD ‘Who came?’)
b. Not only JOHN_F came, John and Mary BOTH_F came.
(QUD: Did [only John came] or [both John and Mary came]?)

Third, even when both components appear to be at-issue, a QUD can be constructed regardless of $\psi_P \Leftarrow \phi_A$ as long as ‘Fluctuation’ is satisfied—namely, the two components are verified by different worlds/ situations. More details are offered below.

7.2. Fluctuation

The ‘also’/‘even’-sentences in (94) and (95) are natural despite that the asserted component entails the presupposed additive component.¹³

- (94) a. Bie shuo liang-ge le, (lian) SAN-ge-ren ye zai-de-xia.
Not say two-CL LE, LIAN three-CL-person YE carry-can-into
‘Needless to say two, [this car] can even carry THREE people.’
b. Zhe-liang-che zai-de-xia liang-ge-ren, ye zai-de-xia SAN-ge-ren.
this-CL-car carry-can-into two-CL-person, YE carry-can-into three-CL-person
‘This car can carry two people, it also can carry THREE people.’
- (95) a. Bie shuo liang-ge le, (lian) SAN-ge-ren ye zai-guo.
Not say two-CL LE, LIAN three-CL-person YE carry-EXP
‘Needless to say two, [this car] even has ever carried THREE people.’
b. Zhe-liang-che zai-guo liang-ge-ren, ye zai-guo SAN-ge-ren.
this-CL-car carry-EXP two-CL-person, YE carry-EXP three-CL-person
‘This car has carried two people, it also has carried THREE people.’

Assuming a QUD-based analysis, I argue that these sentences are acceptable because the asserted components may serve as direct answers to the well-formed QUD in (96). In particular, the modal morpheme *-de* in (94) existentially quantifies over worlds, while the experiential morpheme *-guo* in (95) existentially quantifies over situations.

- (96) QUD: Knowing that this car carries two people in some worlds/ situations, how many people does it carry in some other worlds/ situations?

Crucially, the QUD in (96) is felicitous only when the backgrounded part and the at-issue part are verified by distinct worlds/situations. As a result, the sentences in (94) and (95) cannot be used to describe a scenario where the car carries three people in all relevant worlds/situations, even though it logically entails that this car carries two people in those worlds/situations. This requirement replicates the gist of the *Fluctuation Constraint* (Dayal, 2009), that is, the realized options cannot be kept constant across worlds.¹⁴

¹³Example (94a) is modified from Liu (2023: ex. 50). Observing this pattern, Liu (2023) argues that ‘even’ *ye*, in contrast to ‘also’ *ye*, is not subject to independence. However, this pattern is also attested with ‘also’ *ye*, as seen in (94b). Given this parallel behavior between ‘even’ *ye* and ‘also’ *ye*, I pursue an explanation independent of the ‘also’–‘even’ contrast.

¹⁴There are multiple ways to recast Fluctuation, such as *Viability* (Dayal, 2013), *Revised Viability* (Szabolcsi, 2019), *Relativized Exclusivity* (Xiang, 2022), among others.

The Fluctuation constraint was originally proposed to account for the distribution of FCIs, especially the asymmetry that FCIs are grammatical in ‘can’-sentences but not in ‘must’-sentences or episodic sentences. For details on how Fluctuation affects the distribution of *ye* in sentences with FCIs, see Xiang 2025.

In comparison, the sentences in (97) are unacceptable because they do not have a well-formed QUD—(98) is infelicitous. These sentences involve universal quantification over worlds or situations, yielded by the use of *dei* ‘have to’, *bixu* ‘must’, and *zongshi* ‘always’, which inevitably violates Fluctuation.

- (97) a. # Bie shuo liang-ge-ren le, lian SAN-ge-ren ye dei zai-xia.
Not say two-CL-person LE, LIAN three-CL-person YE have.to carry-into
‘# Needless to say two, (this car) even has to carry THREE people.’
b. # Zhe-liang-che bixu zai liang-ge-ren, ye bixu zai SAN-ge-ren.
this-CL-car must carry two-CL-person, YE must carry three-CL-person
‘# This car must carry two people, it also must carry THREE people.’
c. # Zhe-liang-che zongshi zai liang-ge-ren, ye zongshi zai SAN-ge-ren.
this-CL-car always carry two-CL-person, YE always carry three-CL-person
‘# This car always carries two people, it also always carries THREE people.’
(98) QUD: # Knowing that this car carries two people in all worlds/situations, how many people does it carry in all worlds/situations?

For the same reason, the entailing pattern is unavailable in progressive or simple episodic sentences, since in such contexts the additive component and the asserted component are concerned about a single and the same world/situation, as exemplified in (99) and (100).

- (99) a. # Bie shuo liang-ge-ren le, lian SAN-ge-ren ye zai-zhe/le.
Not say two-CL-person LE, LIAN three-CL-person YE carry-PROG/PERF
‘# Needless to say two, [this car] is even carrying/ even carried THREE people.’
b. # Zhe-liang-che zai-zhe/le liang-ge-ren, ye zai-zhe/le
this-CL-car carry-PROG/PERF two-CL-person, YE carry-PROG/PERF
SAN-ge-ren.
three-CL-person
‘# This car is carrying/ carried two people, it is also carrying/ also carried THREE people.’
(100) QUD: # Knowing that this car carries two people in a prior/ an ongoing situation, how many people does it carry in that situation?

7.3. Violation of independence in minimizer-licensing sentences

Section 4 has argued that in (101), repeated from (37a), the minimizer is interpreted below negation, resulting in an LF where the preadjacent sentence entails all the alternatives.

- (101) Mali (lian) YI-ge-ren dou/ye mei qing.
Mary LIAN one-CL-person DOU/YE NEG invite
‘Mary didn’t even invite a SINGLE person.’

Such cases pose a challenge to traditional approaches to independence, but is predicted by the proposed QUD-based account. I argue that the QUD addressed by the preadjacent sentence

of (37a) is not (102a), which is ill-formed due to a violation of the negative island constraint, but rather (102b) or (102c). The assertion denies the existential presupposition of the QUD, and the alternatives triggered by the minimizer are not at-issue.

- (102) a. * How many people didn't Mary invite?
 b. Who did Mary invite?
 c. How many people did Mary invite?

As predicted, although the two sentences in (103) are logically equivalent, only (103a) can be transformed into a *lian ... ye/dou* constructions, as seen in (104).

- (103) a. Mei YI-ge-ren jiege.
 NEG one-CL-person pass.
 'No one passed.' ($\neg \gg \exists$)
 b. SUOYOU-ren dou mei jige.
 all-person DOU_{ALL} NEG pass
 'EVERYone didn't pass.' ($\forall \gg \neg$)
- (104) a. Lian YI-ge-ren ye/dou mei jige.
 LIAN ONE-CL-person YE/DOU_{EVEN} NEG pass
 'Not even a SINGLE person passed.' ($\text{EVEN} \gg \neg \gg \exists$)
 b. * Lian SUOYOU-ren ye/dou (dou) mei jige.
 LIAN all-person YE/DOU_{EVEN} DOU_{ALL} NEG pass
 '*Even EVERYone didn't pass.' ($\text{*EVEN} \gg \forall \gg \neg$)

8. Conclusions and remarks

This paper investigates the cross-linguistic puzzle that additive expressions frequently develop scalar uses, focusing on the Mandarin additive particle *ye* and comparing it with *dou*, which likewise exhibits a scalar use.

The paper starts by developing a uniform analysis for the 'also' and 'even' uses of *ye*, arguing that all of its presuppositional properties—including focus-sensitivity, additivity, and scalarity—derive from a single definedness condition called *Identity to Anti-Exclusion*. Under this analysis, *ye* is truth-conditionally vacuous, but requires the asserted component of a sentence to be contextually identical to the anti-exclusion of its additive component. Moreover, the 'also'-to-'even' alternation in *ye* results from defining exclusivity in terms of likelihood rather than logical strength, which parallels the distributor-to-'even' alternation of *dou*. The shared scalarity between *dou* and *ye* explains why these two particles display nearly identical behavior in *lian*-sentences, including cases where they associate with a minimizer.

The paper further clarifies the distribution of *ye* and *dou* in more complex scalar environments, especially evaluative statements and scalar concessives. The analysis concludes that, in these two constructions, despite their scalar character, *ye* does not function as a scalar operator 'even', but instead retains its identity as the non-scalar additive operator 'also'. In evaluative contexts, *ye* takes wide scope over the evaluative marker *le*, targeting the evaluative inferences

introduced by *le*. In scalar concessives, *ye* co-occurs with a concessive subordinator because its additive presupposition, realized as an Alternative Normality inference, produces discourse effects parallel to the Abnormality inference contributed by the concessive subordinator. The analysis of scalar concessives also challenges the typological assumption that scalarity is a mechanical necessity for deriving concessivity.

Since the behavior of *ye* in evaluative statements and scalar concessives is largely driven by additivity, it follows that *dou*, as a non-additive operator, exhibits markedly different semantic effects and distributional patterns in these constructions. In evaluative statements, *dou* takes narrow scope below the evaluative marker *le* and functions as a scalar anti-exhaustifier, yielding an anti-exclusion inference over lower-ranked alternatives. In scalar concessives, *dou* in fact associates with a covert unconditional clause, a use that falls under its FC-trigger function.

Finally, this work revisits the traditional independence requirement on additive particles and proposes a QUD-based account. This account explains why *ye* may remain felicitous even in the presence of an entailment relation between the additive and asserted components, provided that Fluctuation is satisfied.

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